

Tool Learning and Function Calling for LLM Agents: A Survey

Tool learning and function calling have become central mechanisms for turning large language models from text predictors into agents that can query information sources, execute programs, operate web and software interfaces, and coordinate multi-step workflows. This survey reviews the rapidly expanding research literature on tool-augmented LLM agents from the perspective of interfaces, learning signals, planning loops, execution environments, evaluation, and safety. We synthesize an arXiv-derived corpus of 300 scholarly papers collected during this run and organize the field into typed function calling, tool discovery, retrieval and memory, code execution, web and embodied action, training and alignment, benchmarks, and governance. The survey argues that tool use is best understood as a joint problem of semantic grounding, state tracking, constrained action generation, feedback interpretation, and risk management. We close by identifying open problems in reliable tool selection, compositional generalization, provenance-aware memory, secure execution, user oversight, and reproducible evaluation.

Keywords:

large language models; tool learning; function calling; agents; retrieval augmentation; evaluation; safety

1 Introduction

Large language models (LLMs) are increasingly deployed as agents rather than as isolated text generators. In an agentic setting, a model receives a goal, maintains a partial state, chooses actions, calls tools, observes external feedback, and iteratively revises its plan. Tool learning and function calling are therefore not peripheral user-interface features; they are the operational layer that connects neural language understanding to symbolic APIs, search engines, program interpreters, databases, web browsers, robotics controllers, and human approval workflows [1, 8, 51, 94, 150, 169, 220, 239]. This shift changes the unit of analysis from a single answer to a trajectory of decisions whose correctness depends on both language understanding and environment dynamics.

The phrase *function calling* usually refers to constrained generation of a typed invocation: a model selects a named function and fills arguments according to a schema. The broader phrase *tool learning*

includes tool discovery, documentation understanding, retrieval of relevant tools, argument synthesis, execution monitoring, response grounding, and adaptation from demonstrations or reinforcement signals [1, 8, 51, 94, 150, 169, 220, 239]. The two ideas meet in modern LLM agents: a function-call interface gives the model an executable action space, while tool-learning methods teach the model how to search, select, sequence, and verify actions in that space.

This survey treats tool-augmented LLM agents as socio-technical systems composed of models, tool specifications, memory, policies, evaluators, and execution sandboxes. The central claim is that progress in this area depends on aligning three representations: natural-language intent, machine-readable tool schemas, and observed state transitions. A brittle mapping among these representations produces familiar failures: invalid arguments, hallucinated APIs, stale retrieved evidence, unsafe shell actions, unrecoverable web navigation, and plausible but unverified final answers. Conversely, robust systems make tool affordances explicit, instrument execution

traces, and use feedback to improve future calls.

2 Background

2.1 From Language Modeling to Agentic Action

Classical language modeling predicts token sequences; agentic LLM systems predict and execute action sequences. This distinction matters because actions have effects outside the transcript. A search query changes the evidence available to the model; a database update changes persistent state; a code execution may produce files, exceptions, or resource use; and a web interaction may trigger irreversible side effects. Tool learning therefore connects natural-language processing with planning, program synthesis, human-computer interaction, reinforcement learning, and software engineering. Recent agent work explores how models decompose tasks, store memory, call tools, and evaluate intermediate observations [1, 8, 51, 94, 150, 169, 220, 239].

2.2 Interfaces, Schemas, and Affordances

A tool interface exposes an affordance: a structured way to change or inspect the world. Function calling reduces the open-endedness of generation by forcing the model to choose a function name and fill typed arguments, but the schema alone does not solve the semantic problem of when a tool is relevant, which arguments are valid, and how to interpret failures. Documentation quality, argument names, examples, return-value conventions, and error messages all influence the agent’s effective policy. Tool-use research consequently studies both model-side adaptation and environment-side interface design [1, 8, 51, 94, 150, 169, 220, 239].

2.3 Retrieval, Memory, and Grounding

Retrieval-augmented generation and agent memory are adjacent to tool use because both provide stateful access to information outside the model parameters. A retrieval call can be understood as a tool whose output must be ranked, compressed, cited, and reconciled with the user’s goal. Long-horizon agents additionally need episodic memory, task state, and provenance records so that later calls can be explained and audited [8, 39, 54, 80, 95, 169, 179, 200]. The boundary between retrieval and tool use is therefore porous: many practical systems combine vector search, web search, code execution, and structured

database operations in one controller loop.

3 Methods

3.1 Search Protocol and Corpus Construction

Following the task requirement to avoid precomputed inventories, the literature corpus for this survey was generated during this run from the arXiv structured API. Queries combined the topic terms with tool-use, function-calling, API, agent, retrieval, planning, code, web, benchmark, training, and safety terms. Records were retained when their title or abstract explicitly connected LLMs or language models with at least one tool, agent, environment, evaluation, or execution concept. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints. The corpus is intentionally broad: tool learning is a system-level topic, so papers on retrieval, code execution, web agents, embodied agents, and safety are included when they illuminate how LLMs act through external interfaces.

3.2 A Taxonomy of Tool-Augmented LLM Agents

We organize the field around six design questions. First, what is the *action space*: a fixed set of typed functions, a retrieved tool library, a shell, a browser, a robotics API, or a mixture of interfaces? Second, how is a tool represented to the model: natural-language documentation, JSON schema, demonstrations, execution traces, or learned embeddings? Third, how does the agent choose and sequence calls: direct prompting, planner-executor decomposition, tree/search-based reasoning, learned policies, or reinforcement learning? Fourth, how are observations grounded: raw tool outputs, summarized evidence, citations, screenshots, exceptions, or state diffs? Fifth, how is success evaluated: single-call exact match, trajectory success, human preference, task completion, cost, latency, and safety constraints? Sixth, how is risk controlled: sandboxing, permission checks, constrained decoding, monitoring, rollback, and human approval.

3.3 Tool Discovery and Selection

Tool discovery addresses the case where the available tool set is too large or dynamic to include in every prompt. A common architecture retrieves candidate tools using names, descriptions, schemas, ex-

amples, or learned embeddings, then asks the model to select one or more functions. The hard part is not only matching words in the user request; it is also recognizing latent preconditions, side effects, required authentication, cost, and compatibility among tools. Strong systems therefore combine retrieval with reranking, argument validation, and post-execution repair. This line of work is represented throughout the retained tool and API literature [1, 8, 51, 94, 150, 169, 220, 239].

3.4 Argument Synthesis and Constrained Generation

Function calling converts language into typed arguments. Compared with free-form prompting, this setting gives developers stronger contracts: enum values, required fields, nested objects, date formats, and admissible ranges can be enforced before execution. Yet argument synthesis remains difficult when the request is underspecified, when defaults are hidden in documentation, or when tool outputs from earlier steps must be transformed into later inputs. Methods in this family use schema-aware prompting, examples, constrained decoding, self-correction, and validation loops. A useful abstraction is to treat every call as a small program whose type checks, runtime behavior, and provenance should be visible to both the model and the user.

3.5 Planning, Decomposition, and Execution Loops

Long tasks require an agent to transform a user goal into a plan, execute calls, observe results, and revise. The planning literature contributes decomposition, search, reflection, and state tracking; tool-learning work contributes executable actions and feedback. ReAct-style interleaving of reasoning and acting is one influential pattern, but many variants separate planning from execution, maintain explicit task graphs, or use verifiers to decide when to stop. The retained agent-planning papers study these choices across textual, web, software, and embodied environments [1, 8, 51, 94, 150, 169, 220, 239].

3.6 Code Execution and Program-Aided Tool Use

Code interpreters are unusually powerful tools because they give LLMs access to loops, data structures, numerical libraries, file systems, and tests. Program-aided approaches ask the model to write code that computes an answer, calls an API, transforms data, or validates an intermediate result. This

makes the boundary between tool use and program synthesis thin: a model may call a fixed function, generate a new helper function, or compose both inside a script. Code-oriented agent research also provides rigorous feedback through compiler errors, unit tests, execution traces, and repository-level benchmarks [51, 80, 131, 147, 169, 207, 220, 276].

3.7 Web, GUI, and Embodied Agents

Web and GUI agents expose noisy, partially observable environments rather than clean function signatures. The model must parse pages or screenshots, choose clicks and text inputs, wait for state changes, and recover from navigation errors. Embodied agents add perception, spatial memory, low-level control, and physical safety. These settings stress-test the assumption that tools have simple schemas: many real tools are interactive, visual, stateful, and asynchronous. Research on web automation, computer control, robotics, and multimodal agents expands the tool-learning problem from API selection to closed-loop control [8, 105, 133, 169, 178, 200, 239, 252].

3.8 Training Signals and Policy Improvement

Tool use can be induced by prompting alone, but robust behavior usually benefits from demonstrations, synthetic traces, self-generated data, preference optimization, reinforcement learning, or verifier-guided search. Training data may consist of successful call sequences, failed attempts with repairs, human-labeled tool choices, or environment rewards. The most important design choice is the feedback granularity: outcome-only rewards encourage task completion but may hide invalid intermediate behavior, while step-level supervision can teach schemas and recovery but is expensive to collect. Recent training and alignment studies investigate this trade-off for LLM agents and tool-mediated reasoning [39, 54, 94, 131, 147, 150, 239, 299].

3.9 Benchmarks and Evaluation

Evaluation has moved from single-turn question answering toward trajectory-level task completion. A benchmark may score whether the final answer is correct, whether the right tools were used, whether each argument was valid, whether the execution trace is efficient, and whether the system respected safety constraints. Reproducibility is difficult because tools, websites, documents, and model APIs change over time. Benchmarks therefore need ver-

sioned environments, stable oracles, cost accounting, and logs that preserve the full interaction. The retained evaluation literature reflects this shift from static accuracy to interactive reliability [1, 8, 51, 131, 150, 169, 220, 239].

3.10 Representative Literature Map

The following map cites the full retained corpus by theme. It is not intended as a flat reading list; rather, it makes the survey’s coverage explicit and shows how papers were assigned to methodological families.

Tool interfaces and function calling The tool interfaces and function calling cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [1, 51, 131, 147, 150, 179, 205, 220, 239]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [24, 42, 54, 73, 75, 207, 251, 276, 299]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [32, 110, 121, 130, 148, 157, 166, 175, 189]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [77, 90, 103, 114, 136, 193, 197, 291, 300]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [13, 28, 86, 96, 102, 192, 209, 224, 269]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [27, 55, 65, 98, 158, 202, 225, 246, 266]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [16, 41, 58, 70, 143, 145, 199, 267, 281]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [7, 15, 21, 36, 135, 139, 163, 177, 282]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [14, 38, 49, 128, 160, 222, 241, 264, 288]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [46, 53, 68, 89, 106, 162, 181, 188, 278]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [22, 30, 43, 118, 146, 149, 187, 218, 256]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [2, 5, 74, 79, 85, 111, 137, 268, 289]. Additional retained studies broaden the

cluster with variants of interfaces, domains, training signals, and deployment assumptions [18, 167, 176].

Agent planning and reasoning The agent planning and reasoning cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [8, 39, 80, 94, 133, 169, 200, 237, 252]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [66, 67, 95, 105, 221, 240, 261, 277, 284]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [6, 20, 26, 44, 64, 69, 120, 134, 182]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [17, 29, 108, 144, 232, 236, 260, 293, 298]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [72, 115, 165, 186, 201, 211, 228, 272, 285]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [37, 60, 117, 124, 153, 159, 219, 258, 290]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [3, 87, 122, 156, 198, 234, 245, 254, 255]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [23, 34, 40, 59, 97, 109, 173, 226, 273]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [12, 31, 33, 119, 125, 170, 183, 194, 253]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [91, 107, 155, 174, 190, 214, 244, 274, 286]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [25, 52, 57, 81, 82, 141, 216, 257, 287]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [4, 10, 50, 61, 196, 235, 242, 280, 295]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [11, 71, 83, 113, 116, 203, 204, 229, 283]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [88, 101, 112, 126, 151, 213, 231].

Retrieval, memory, and knowledge access The retrieval, memory, and knowledge access cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [62, 84, 99, 100, 123, 185, 217, 243, 259].

Code, program execution, and software tools The code, program execution, and software tools cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [48, 180, 270].

Web, gui, and embodied environments The web, gui, and embodied environments cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [172, 227, 247, 294].

Training, alignment, and reinforcement learning The training, alignment, and reinforcement learning cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [47, 56, 78, 93, 154, 164, 248, 249, 262]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [138, 250, 265].

Benchmarks, evaluation, and datasets The benchmarks, evaluation, and datasets cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [45, 129, 140, 178, 195, 206, 208, 210, 292]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [9, 19, 132, 161, 171, 184, 215, 223, 230]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [35, 76, 142, 168, 212, 271, 275, 279, 296]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [63, 92, 104, 127, 152, 191, 233].

Safety, reliability, and governance The safety, reliability, and governance cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [238, 263].

Multi-agent coordination and collaboration The multi-agent coordination and collaboration cluster contains work that treats LLMs as controllers of

typed resources, stateful environments, or evaluation tasks [297].

4 Challenges and Prospects

4.1 Reliability Under Distribution Shift

Tool-using agents often fail when schemas, websites, documents, or APIs differ from the examples in their prompts or training traces. Distribution shift is amplified by tool composition: a small extraction error can become an invalid argument, which then produces an exception, which then leads the model to invent a recovery path. Future work needs calibrated uncertainty over tool choice, richer validation before execution, and agents that can ask clarifying questions when the action space is under-specified.

4.2 Semantic Grounding and Provenance

Function calls should be grounded in user intent and external evidence. This requires provenance records that connect each argument to a phrase in the user request, a retrieved document, a previous observation, or a model inference. Without such records, an agent may appear competent while silently mixing stale memory, hallucinated facts, and real tool outputs. Grounded memory and retrieval methods offer partial solutions, but tool-rich systems need provenance that survives multi-step transformations and can be inspected after deployment [8, 39, 54, 80, 95, 169, 179, 200].

4.3 Safety, Permissions, and Secure Execution

Tools increase capability and risk at the same time. A model that can browse, write files, send messages, or execute code can also leak data, follow malicious instructions, exceed user intent, or damage persistent state. Safety mechanisms must therefore operate at several layers: tool documentation, permission scopes, sandboxing, argument filters, monitors, rollback, and human approval for irreversible actions. Security-oriented studies emphasize that prompt injection and untrusted observations are not edge cases but ordinary conditions for web- and document-facing agents [20, 66, 80, 120, 169, 182, 251, 252].

4.4 Cost, Latency, and Systems Design

Agent loops can be expensive because every step may require model inference, retrieval, tool execution, validation, and logging. A practical system must decide when to call a tool, when to rely on model knowledge, when to summarize context, and when to stop. Future agents will likely use smaller controllers, cached tool embeddings, learned stopping policies, and asynchronous execution. The research opportunity is to treat tool use not only as a reasoning problem but also as a systems problem with resource constraints.

4.5 Toward Compositional and Auditable Agents

The long-term prospect is an agent architecture in which tools are typed, discoverable, composable, and auditable. Such an agent would select tools based on formal affordances and learned experience, generate arguments with type and provenance constraints, execute in sandboxed environments, and produce final answers backed by traceable observations. Progress will depend on shared benchmarks, open execution traces, and evaluation protocols that reward reliable behavior rather than only persuasive final text.

5 Conclusion

Tool learning and function calling are reshaping LLM research by giving language models executable interfaces to external systems. This survey reviewed the field through the lenses of interfaces, tool discovery, argument synthesis, planning, retrieval, code execution, web and embodied environments, training, evaluation, and safety. The central lesson is that tool use is not a single capability; it is a stack of representations and control decisions that must remain aligned over time. The next generation of LLM agents will require stronger schemas, better feedback, verifiable provenance, secure execution, and benchmarks that measure entire trajectories. When these pieces mature together, function calling can become less like a prompt-engineering trick and more like a principled substrate for reliable AI systems.

References

- [1] Ibrahim Abdelaziz, Kinjal Basu, Mayank Agarwal, Sadhana Kumaravel, Matthew Stallone, Rameswar Panda, Yara Rizk, GP Bhargav, Maxwell Crouse, Chulaka Gunasekara, Shajith Ikbali, Sachin Joshi, Hima Karanam, Vineet Kumar, Asim Munawar, Sumit Neelam, Dinesh Raghu, Udit Sharma, Adriana Meza Soria, Dheeraj Sreedhar, et al. Granite-Function Calling Model: Introducing Function Calling Abilities via Multi-task Learning of Granular Tasks. *arXiv preprint arXiv:2407.00121*, 2024. URL <https://arxiv.org/abs/2407.00121>.
- [2] Mayank Agarwal, Ibrahim Abdelaziz, Kinjal Basu, Merve Unuvar, Luis A. Lastras, Yara Rizk, and Pavan Kapanipathi. ToolRM: Outcome Reward Models for Tool-Calling Large Language Models. *arXiv preprint arXiv:2509.11963*, 2025. URL <https://arxiv.org/abs/2509.11963>.
- [3] Saaket Agashe, Yue Fan, Anthony Reyna, and Xin Eric Wang. LLM-Coordination: Evaluating and Analyzing Multi-agent Coordination Abilities in Large Language Models. *arXiv preprint arXiv:2310.03903*, 2023. URL <https://arxiv.org/abs/2310.03903>.
- [4] Gokturk Aytug Akarlar. Beyond Prompt Engineering: Neuro-Symbolic-Causal Architecture for Robust Multi-Objective AI Agents. *arXiv preprint arXiv:2510.23682*, 2025. URL <https://arxiv.org/abs/2510.23682>.
- [5] Lisa Alazraki and Marek Rei. Meta-Reasoning Improves Tool Use in Large Language Models. *arXiv preprint arXiv:2411.04535*, 2024. URL <https://arxiv.org/abs/2411.04535>.
- [6] Samuel Onimpa Alfred and Veera Sundararaghavan. From Data to Theory: Autonomous Large Language Model Agents for Materials Science. *arXiv preprint arXiv:2604.19789*, 2026. URL <https://arxiv.org/abs/2604.19789>.
- [7] Tamer Alkhoul, Katerina Margatina, James Gung, Raphael Shu, Claudia Zaghi, Monica Sunkara, and Yi Zhang. CONFETTI: Conversational Function-Calling Evaluation Through Turn-Level Interactions. *arXiv preprint arXiv:2506.01859*, 2025. URL <https://arxiv.org/abs/2506.01859>.
- [8] Rahul Suresh Babu and Adarsh Agrawal. Self-Healing Agentic Orchestrators for Reliable Tool-Augmented Large Language Model Systems. *arXiv preprint arXiv:2606.01416*, 2026. URL <https://arxiv.org/abs/2606.01416>.
- [9] Liam Barkley and Brink van der Merwe. Investigating the Role of Prompting and External Tools in Hallucination Rates of Large Language Models. *arXiv preprint arXiv:2410.19385*, 2024. URL <https://arxiv.org/abs/2410.19385>.
- [10] Daniele Barolo. From Tokens To Agents: A Researcher’s Guide To Understanding Large Language Models. *arXiv preprint arXiv:2603.19269*, 2026. URL <https://arxiv.org/abs/2603.19269>.
- [11] Jonas Becker. Multi-Agent Large Language Models for Conversational Task-Solving. *arXiv preprint*

- arXiv:2410.22932*, 2024. URL <https://arxiv.org/abs/2410.22932>.
- [12] Jonas Becker, Lars Benedikt Kaesberg, Niklas Bauer, Jan Philip Wahle, Terry Ruas, and Bela Gipp. MALLM: Multi-Agent Large Language Models Framework. *arXiv preprint arXiv:2509.11656*, 2025. URL <https://arxiv.org/abs/2509.11656>.
 - [13] Vibha Belavadi, Tushar Vatsa, Dewang Sultania, Suhas Suresha, Ishita Verma, Cheng Chen, Tracy Holloway King, and Michael Friedrich. RouteNator: A Router-Based Multi-Modal Architecture for Generating Synthetic Training Data for Function Calling LLMs. *arXiv preprint arXiv:2505.10495*, 2025. URL <https://arxiv.org/abs/2505.10495>.
 - [14] Yannis Belkhit, Giulio Zizzo, Sergio Maffei, Seshu Tirupathi, and John D. Kelleher. Breaking MCP with Function Hijacking Attacks: Novel Threats for Function Calling and Agentic Models. *arXiv preprint arXiv:2604.20994*, 2026. URL <https://arxiv.org/abs/2604.20994>.
 - [15] Nirav Bhan, Shival Gupta, Sai Manaswini, Ritik Baba, Narun Yadav, Hillori Desai, Yash Choudhary, Aman Pawar, Sarthak Shrivastava, and Sudipta Biswas. Benchmarking Floworks against OpenAI & Anthropic: A Novel Framework for Enhanced LLM Function Calling. *arXiv preprint arXiv:2410.17950*, 2024. URL <https://arxiv.org/abs/2410.17950>.
 - [16] Thierry Blankenstein, Jialin Yu, Zixuan Li, Vasilis Plachouras, Sunando Sengupta, Philip Torr, Yarin Gal, Alasdair Paren, and Adel Bibi. Bias-Busters: Uncovering and Mitigating Tool Selection Bias in Large Language Models. *arXiv preprint arXiv:2510.00307*, 2025. URL <https://arxiv.org/abs/2510.00307>.
 - [17] Hongru Cai, Yongqi Li, Wenjie Wang, Fengbin Zhu, Xiaoyu Shen, Wenjie Li, and Tat-Seng Chua. Large Language Models Empowered Personalized Web Agents. *arXiv preprint arXiv:2410.17236*, 2024. URL <https://arxiv.org/abs/2410.17236>.
 - [18] Tianle Cai, Xuezhi Wang, Tengyu Ma, Xinyun Chen, and Denny Zhou. Large Language Models as Tool Makers. *arXiv preprint arXiv:2305.17126*, 2023. URL <https://arxiv.org/abs/2305.17126>.
 - [19] Chen Chen, Kim Young Il, Yuan Yang, Wenhao Su, Yilin Zhang, Xueluan Gong, Qian Wang, Yongsun Zheng, Ziyao Liu, and Kwok-Yan Lam. The Shadow Self: Intrinsic Value Misalignment in Large Language Model Agents. *arXiv preprint arXiv:2601.17344*, 2026. URL <https://arxiv.org/abs/2601.17344>.
 - [20] Fengchao Chen, Tingmin Wu, Van Nguyen, and Carsten Rudolph. Agents at Risk: How Users Unwittingly Undermine LLM Safety. *arXiv preprint arXiv:2601.10758*, 2026. URL <https://arxiv.org/abs/2601.10758>.
 - [21] Guoxin Chen, Zhong Zhang, Xin Cong, Fangda Guo, Yesai Wu, Yankai Lin, Wenzheng Feng, and Yasheng Wang. Learning Evolving Tools for Large Language Models. *arXiv preprint arXiv:2410.06617*, 2024. URL <https://arxiv.org/abs/2410.06617>.
 - [22] Haotian Chen, Zijun Song, Boye Niu, Ke Zhang, Litu Ou, Yaxi Lu, Zhong Zhang, Xin Cong, Yankai Lin, Zhiyuan Liu, and Maosong Sun. ToLeaP: Rethinking Development of Tool Learning with Large Language Models. *arXiv preprint arXiv:2505.11833*, 2025. URL <https://arxiv.org/abs/2505.11833>.
 - [23] Huaben Chen, Wenkang Ji, Lufeng Xu, and Shiyu Zhao. Multi-Agent Consensus Seeking via Large Language Models. *arXiv preprint arXiv:2310.20151*, 2023. URL <https://arxiv.org/abs/2310.20151>.
 - [24] Jiaao Chen, Jingyuan Qi, Mingye Gao, Wei-Chen Wang, Hanrui Wang, and Di Jin. EigenData: A Self-Evolving Multi-Agent Platform for Function-Calling Data Synthesis, Auditing, and Repair. *arXiv preprint arXiv:2603.05553*, 2026. URL <https://arxiv.org/abs/2603.05553>.
 - [25] Jiaqi Chen, Yuxian Jiang, Jiachen Lu, and Li Zhang. S-Agents: Self-organizing Agents in Open-ended Environments. *arXiv preprint arXiv:2402.04578*, 2024. URL <https://arxiv.org/abs/2402.04578>.
 - [26] Junzhe Chen, Xuming Hu, Shuodi Liu, Shiyu Huang, Wei-Wei Tu, Zhaofeng He, and Lijie Wen. LLMarena: Assessing Capabilities of Large Language Models in Dynamic Multi-Agent Environments. *arXiv preprint arXiv:2402.16499*, 2024. URL <https://arxiv.org/abs/2402.16499>.
 - [27] Mingyang Chen, Haoze Sun, Tianpeng Li, Fan Yang, Hao Liang, Keer Lu, Bin Cui, Wentao Zhang, Zenan Zhou, and Weipeng Chen. Facilitating Multi-turn Function Calling for LLMs via Compositional Instruction Tuning. *arXiv preprint arXiv:2410.12952*, 2024. URL <https://arxiv.org/abs/2410.12952>.
 - [28] Wei Chen, Zhiyuan Li, and Mingyuan Ma. Octopus: On-device language model for function calling of software APIs. *arXiv preprint arXiv:2404.01549*, 2024. URL <https://arxiv.org/abs/2404.01549>.
 - [29] Weizhe Chen, Sven Koenig, and Bistra Dilikina. RePrompt: Planning by Automatic Prompt Engineering for Large Language Models Agents. *arXiv preprint arXiv:2406.11132*, 2024. URL <https://arxiv.org/abs/2406.11132>.
 - [30] Wenjie Chen, Wenbin Li, Di Yao, Xuying Meng,

- Chang Gong, and Jingping Bi. GTool: Graph Enhanced Tool Planning with Large Language Model. *arXiv preprint arXiv:2508.12725*, 2025. URL <https://arxiv.org/abs/2508.12725>.
- [31] Yi-Chang Chen, Po-Chun Hsu, Chan-Jan Hsu, and Da shan Shiu. Enhancing Function-Calling Capabilities in LLMs: Strategies for Prompt Formats, Data Integration, and Multilingual Translation. *arXiv preprint arXiv:2412.01130*, 2024. URL <https://arxiv.org/abs/2412.01130>.
- [32] Yifei Chen, Sarra Habchi, and Lili Wei. MIMIC-Py: An Extensible Tool for Personality-Driven Automated Game Testing with Large Language Models. *arXiv preprint arXiv:2604.07752*, 2026. URL <https://arxiv.org/abs/2604.07752>.
- [33] Zehui Chen, Weihua Du, Wenwei Zhang, Kuikun Liu, Jiangning Liu, Miao Zheng, Jingming Zhuo, Songyang Zhang, Dahua Lin, Kai Chen, and Feng Zhao. T-Eval: Evaluating the Tool Utilization Capability of Large Language Models Step by Step. *arXiv preprint arXiv:2312.14033*, 2023. URL <https://arxiv.org/abs/2312.14033>.
- [34] Zehui Chen, Kuikun Liu, Qiuchen Wang, Wenwei Zhang, Jiangning Liu, Dahua Lin, Kai Chen, and Feng Zhao. Agent-FLAN: Designing Data and Methods of Effective Agent Tuning for Large Language Models. *arXiv preprint arXiv:2403.12881*, 2024. URL <https://arxiv.org/abs/2403.12881>.
- [35] Zhipeng Chen, Kun Zhou, Beichen Zhang, Zheng Gong, Wayne Xin Zhao, and Ji-Rong Wen. Chat-CoT: Tool-Augmented Chain-of-Thought Reasoning on Chat-based Large Language Models. *arXiv preprint arXiv:2305.14323*, 2023. URL <https://arxiv.org/abs/2305.14323>.
- [36] Aijia Cheng, Kailong Wang, Ling Shi, and Yongxin Zhao. R2IF: Aligning Reasoning with Decisions via Composite Rewards for Interpretable LLM Function Calling. *arXiv preprint arXiv:2604.20316*, 2026. URL <https://arxiv.org/abs/2604.20316>.
- [37] Mingyue Cheng, Shuo Yu, Daoyu Wang, Qingchuan Li, Xiaoyu Tao, Jie Ouyang, Yucong Luo, Yitong Zhou, Qi Liu, and Enhong Chen. Agent-R1: A Unified and Modular Framework for Agentic Reinforcement Learning. *arXiv preprint arXiv:2511.14460*, 2025. URL <https://arxiv.org/abs/2511.14460>.
- [38] Zihao Cheng, Hongru Wang, Zeming Liu, Yuhang Guo, Yuanfang Guo, Yunhong Wang, and Haifeng Wang. ToolSpectrum : Towards Personalized Tool Utilization for Large Language Models. *arXiv preprint arXiv:2505.13176*, 2025. URL <https://arxiv.org/abs/2505.13176>.
- [39] Sadia Sultana Chowa, Riasad Alvi, Subhey Sadi Rahman, Md Abdur Rahman, Mohaimenul Azam Khan Raiaan, Md Rafiqul Islam, Mukhtar Hussain, and Sami Azam. From Language to Action: A Review of Large Language Models as Autonomous Agents and Tool Users. *arXiv preprint arXiv:2508.17281*, 2025. URL <https://arxiv.org/abs/2508.17281>.
- [40] Zhixuan Chu, Yan Wang, Feng Zhu, Lu Yu, Longfei Li, and Jinjie Gu. Professional Agents – Evolving Large Language Models into Autonomous Experts with Human-Level Competencies. *arXiv preprint arXiv:2402.03628*, 2024. URL <https://arxiv.org/abs/2402.03628>.
- [41] Luca Collini, Siddharth Garg, and Ramesh Karri. C2HLSC: Leveraging Large Language Models to Bridge the Software-to-Hardware Design Gap. *arXiv preprint arXiv:2412.00214*, 2024. URL <https://arxiv.org/abs/2412.00214>.
- [42] Sijia Cui, Aiyao He, Shuai Xu, Hongming Zhang, Yanna Wang, Qingyang Zhang, Yajing Wang, and Bo Xu. Self-Guided Function Calling in Large Language Models via Stepwise Experience Recall. *arXiv preprint arXiv:2508.15214*, 2025. URL <https://arxiv.org/abs/2508.15214>.
- [43] Yue Cui, Liuyi Yao, Shuchang Tao, Weijie Shi, Yaliang Li, Bolin Ding, and Xiaofang Zhou. Enhancing Tool Learning in Large Language Models with Hierarchical Error Checklists. *arXiv preprint arXiv:2506.00042*, 2025. URL <https://arxiv.org/abs/2506.00042>.
- [44] Hy Dang, Tianyi Liu, Zhuofeng Wu, Jingfeng Yang, Haoming Jiang, Tao Yang, Pei Chen, Zhengyang Wang, Helen Wang, Huasheng Li, Bing Yin, and Meng Jiang. Improving Large Language Models Function Calling and Interpretability via Guided-Structured Templates. *arXiv preprint arXiv:2509.18076*, 2025. URL <https://arxiv.org/abs/2509.18076>.
- [45] Debrup Das, Debopriyo Banerjee, Somak Aditya, and Ashish Kulkarni. MATHSENSEI: A Tool-Augmented Large Language Model for Mathematical Reasoning. *arXiv preprint arXiv:2402.17231*, 2024. URL <https://arxiv.org/abs/2402.17231>.
- [46] Mishca de Costa, Muhammad Anwar, Dave Mercier, Mark Randall, and Issam Hammad. Enhancing Accuracy and Maintainability in Nuclear Plant Data Retrieval: A Function-Calling LLM Approach Over NL-to-SQL. *arXiv preprint arXiv:2506.08757*, 2025. URL <https://arxiv.org/abs/2506.08757>.
- [47] Boyi Deng, Xu Wang, Yaoning Wang, Yu Wan, Yubo Ma, Baosong Yang, Haoran Wei, Jialong Tang, Huan Lin, Ruize Gao, Tianhao Li, Qian Cao, Xuancheng Ren, Xiaodong Deng, An Yang, Fei Huang, Dayiheng Liu, and Jingren Zhou. Qwen-Scope: Turning Sparse Features into Development Tools for Large Language Models. *arXiv preprint*

- arXiv:2605.11887*, 2026. URL <https://arxiv.org/abs/2605.11887>.
- [48] Hanxing Ding, Shuchang Tao, Liang Pang, Zihao Wei, Jinyang Gao, Bolin Ding, Huawei Shen, and Xueqi Cheng. ToolCoder: A Systematic Code-Empowered Tool Learning Framework for Large Language Models. *arXiv preprint arXiv:2502.11404*, 2025. URL <https://arxiv.org/abs/2502.11404>.
 - [49] Peng Ding and Rick Stevens. Unified Tool Integration for LLMs: A Protocol-Agnostic Approach to Function Calling. *arXiv preprint arXiv:2508.02979*, 2025. URL <https://arxiv.org/abs/2508.02979>.
 - [50] Yixin Dong, Charlie F. Ruan, Yaxing Cai, Ruihang Lai, Ziyi Xu, Yilong Zhao, and Tianqi Chen. XGrammar: Flexible and Efficient Structured Generation Engine for Large Language Models. *arXiv preprint arXiv:2411.15100*, 2024. URL <https://arxiv.org/abs/2411.15100>.
 - [51] Yu Du, Fangyun Wei, and Hongyang Zhang. AnyTool: Self-Reflective, Hierarchical Agents for Large-Scale API Calls. *arXiv preprint arXiv:2402.04253*, 2024. URL <https://arxiv.org/abs/2402.04253>.
 - [52] Yifan Duan, Yihong Tang, Xuefeng Bai, Kehai Chen, Juntao Li, and Min Zhang. The Power of Personality: A Human Simulation Perspective to Investigate Large Language Model Agents. *arXiv preprint arXiv:2502.20859*, 2025. URL <https://arxiv.org/abs/2502.20859>.
 - [53] Zhihua Duan and Jialin Wang. Multi-tool Integration Application for Math Reasoning Using Large Language Model. *arXiv preprint arXiv:2408.12148*, 2024. URL <https://arxiv.org/abs/2408.12148>.
 - [54] Lutfi Eren Erdogan, Nicholas Lee, Siddharth Jha, Sehoon Kim, Ryan Tabrizi, Suhong Moon, Coleman Hooper, Gopala Anumanchipalli, Kurt Keutzer, and Amir Gholami. TinyAgent: Function Calling at the Edge. *arXiv preprint arXiv:2409.00608*, 2024. URL <https://arxiv.org/abs/2409.00608>.
 - [55] Bowen Fang, Wen Ye, Yunyue Su, Jinghao Zhang, Qiang Liu, Yesheng Liu, Xin Sun, Shu Wu, Jiabing Yang, Baole Wei, and Liang Wang. ToolWeaver: Weaving Collaborative Semantics for Scalable Tool Use in Large Language Models. *arXiv preprint arXiv:2601.21947*, 2026. URL <https://arxiv.org/abs/2601.21947>.
 - [56] Andrew Zhuoer Feng, Cunxiang Wang, Bosi Wen, Yidong Wang, Yu Luo, Hongning Wang, and Minlie Huang. RLAR: An Agentic Reward System for Multi-task Reinforcement Learning on Large Language Models. *arXiv preprint arXiv:2603.00724*, 2026. URL <https://arxiv.org/abs/2603.00724>.
 - [57] Xiachong Feng, Longxu Dou, Ella Li, Qinghao Wang, Haochuan Wang, Yu Guo, Chang Ma, and Lingpeng Kong. A Survey on Large Language Model-Based Social Agents in Game-Theoretic Scenarios. *arXiv preprint arXiv:2412.03920*, 2024. URL <https://arxiv.org/abs/2412.03920>.
 - [58] Mina Foosherian, Hendrik Purwins, Purna Rathnayake, Touhidul Alam, Rui Teimao, and Klaus-Dieter Thoben. Enhancing Pipeline-Based Conversational Agents with Large Language Models. *arXiv preprint arXiv:2309.03748*, 2023. URL <https://arxiv.org/abs/2309.03748>.
 - [59] Ivar Frisch and Mario Giulianelli. LLM Agents in Interaction: Measuring Personality Consistency and Linguistic Alignment in Interacting Populations of Large Language Models. *arXiv preprint arXiv:2402.02896*, 2024. URL <https://arxiv.org/abs/2402.02896>.
 - [60] Victor Gallego. Discovering Agentic Safety Specifications from 1-Bit Danger Signals. *arXiv preprint arXiv:2604.23210*, 2026. URL <https://arxiv.org/abs/2604.23210>.
 - [61] Chen Gao, Xiaochong Lan, Nian Li, Yuan Yuan, Jingtao Ding, Zhilun Zhou, Fengli Xu, and Yong Li. Large Language Models Empowered Agent-based Modeling and Simulation: A Survey and Perspectives. *arXiv preprint arXiv:2312.11970*, 2023. URL <https://arxiv.org/abs/2312.11970>.
 - [62] Hang Gao and Yongfeng Zhang. INMS: Memory Sharing for Large Language Model based Agents. *arXiv preprint arXiv:2404.09982*, 2024. URL <https://arxiv.org/abs/2404.09982>.
 - [63] Hang Gao and Yongfeng Zhang. Task-Aligned Tool Recommendation for Large Language Models. *arXiv preprint arXiv:2411.09613*, 2024. URL <https://arxiv.org/abs/2411.09613>.
 - [64] Shen Gao, Yuntao Wen, Minghang Zhu, Jianing Wei, Yuhang Cheng, Qunzi Zhang, and Shuo Shang. Simulating Financial Market via Large Language Model based Agents. *arXiv preprint arXiv:2406.19966*, 2024. URL <https://arxiv.org/abs/2406.19966>.
 - [65] Xin Gao, Ruiyi Zhang, Meixi Du, Peijia Qin, and Pengtao Xie. BioTool: A Comprehensive Tool-Calling Dataset for Enhancing Biomedical Capabilities of Large Language Models. *arXiv preprint arXiv:2605.05758*, 2026. URL <https://arxiv.org/abs/2605.05758>.
 - [66] Tarek Gasmi, Ramzi Guesmi, Ines Belhadj, and Jihene Bennaceur. Bridging AI and Software Security: A Comparative Vulnerability Assessment of LLM Agent Deployment Paradigms. *arXiv preprint arXiv:2507.06323*, 2025. URL <https://arxiv.org/abs/2507.06323>.

- [67] Subha Ghoshal and Ali Al-Bustami. When Do Tools and Planning Help Large Language Models Think? A Cost- and Latency-Aware Benchmark. *arXiv preprint arXiv:2601.02663*, 2026. URL <https://arxiv.org/abs/2601.02663>.
- [68] In Gim, Seung seob Lee, and Lin Zhong. Asynchronous LLM Function Calling. *arXiv preprint arXiv:2412.07017*, 2024. URL <https://arxiv.org/abs/2412.07017>.
- [69] Dihong Gong, Pu Lu, Zelong Wang, Meng Zhou, and Xiuqiang He. Training Agents with Weakly Supervised Feedback from Large Language Models. *arXiv preprint arXiv:2411.19547*, 2024. URL <https://arxiv.org/abs/2411.19547>.
- [70] Jian Gong, Youwei Huang, Bo Yuan, Ming Zhu, Zhou Liao, Jianhang Liang, Juncheng Zhan, Jinke Wang, Hang Shu, Mingyue Xiong, Yanjun Ye, Yufan Zu, Yang Zhou, Yihan Ding, Xuannian Chen, Xingyu Lu, Runjie Ban, Bingchao Huang, and Fusen Liu. GhostShell: Streaming LLM Function Calls for Concurrent Embodied Programming. *arXiv preprint arXiv:2508.05298*, 2025. URL <https://arxiv.org/abs/2508.05298>.
- [71] Yanchu Guan, Dong Wang, Yan Wang, Haiqing Wang, Renen Sun, Chenyi Zhuang, Jinjie Gu, and Zhixuan Chu. Explainable Behavior Cloning: Teaching Large Language Model Agents through Learning by Demonstration. *arXiv preprint arXiv:2410.22916*, 2024. URL <https://arxiv.org/abs/2410.22916>.
- [72] Taicheng Guo, Xiuying Chen, Yaqi Wang, Ruidi Chang, Shichao Pei, Nitesh V. Chawla, Olaf Wiest, and Xiangliang Zhang. Large Language Model based Multi-Agents: A Survey of Progress and Challenges. *arXiv preprint arXiv:2402.01680*, 2024. URL <https://arxiv.org/abs/2402.01680>.
- [73] Zhicheng Guo, Sijie Cheng, Hao Wang, Shihao Liang, Yujia Qin, Peng Li, Zhiyuan Liu, Maosong Sun, and Yang Liu. StableToolBench: Towards Stable Large-Scale Benchmarking on Tool Learning of Large Language Models. *arXiv preprint arXiv:2403.07714*, 2024. URL <https://arxiv.org/abs/2403.07714>.
- [74] Piyush Gupta, Sangjae Bae, and David Isele. Graph-Grounded LLMs: Leveraging Graphical Function Calling to Minimize LLM Hallucinations. *arXiv preprint arXiv:2503.10941*, 2025. URL <https://arxiv.org/abs/2503.10941>.
- [75] Skyler Hallinan, Thejas Venkatesh, Xiang Ren, Sai Praneeth Karimireddy, Ashwin Paranjape, Yuhao Zhang, and Jack Hessel. OpaqueToolsBench: Learning Nuances of Tool Behavior Through Interaction. *arXiv preprint arXiv:2602.15197*, 2026. URL <https://arxiv.org/abs/2602.15197>.
- [76] Han Han, Tong Zhu, Xiang Zhang, Mengsong Wu, Hao Xiong, and Wenliang Chen. NesTools: A Dataset for Evaluating Nested Tool Learning Abilities of Large Language Models. *arXiv preprint arXiv:2410.11805*, 2024. URL <https://arxiv.org/abs/2410.11805>.
- [77] Kairong Han, Kun Kuang, Ziyu Zhao, Junjian Ye, and Fei Wu. Causal Agent based on Large Language Model. *arXiv preprint arXiv:2408.06849*, 2024. URL <https://arxiv.org/abs/2408.06849>.
- [78] Bingguang Hao, Zengzhuang Xu, Maolin Wang, Yuntao Wen, Yicheng Chen, Cunyin Peng, Long Chen, Dong Wang, Xiangyu Zhao, Jinjie Gu, Chenyi Zhuang, and Ji Zhang. Reasoning through Exploration: A Reinforcement Learning Framework for Robust Function Calling. *arXiv preprint arXiv:2508.05118*, 2025. URL <https://arxiv.org/abs/2508.05118>.
- [79] Yupu Hao, Pengfei Cao, Zhuoran Jin, Huanxuan Liao, Yubo Chen, Kang Liu, and Jun Zhao. CITI: Enhancing Tool Utilizing Ability in Large Language Models without Sacrificing General Performance. *arXiv preprint arXiv:2409.13202*, 2024. URL <https://arxiv.org/abs/2409.13202>.
- [80] Hadi Hasan, Safaa Salman, Adam Tai Abou Dargham, Ammar Mohanna, and Ali Chehab. Toward Trustworthy Large Language Model Agents in Healthcare. *arXiv preprint arXiv:2607.05055*, 2026. URL <https://arxiv.org/abs/2607.05055>.
- [81] Masum Hasan, Cengiz Ozel, Sammy Potter, and Ehsan Hoque. SAPIEN: Affective Virtual Agents Powered by Large Language Models. *arXiv preprint arXiv:2308.03022*, 2023. URL <https://arxiv.org/abs/2308.03022>.
- [82] Xin He, Liangliang You, Hongduan Tian, Bo Han, Ivor Tsang, and Yew-Soon Ong. Lang-PINN: From Language to Physics-Informed Neural Networks via a Multi-Agent Framework. *arXiv preprint arXiv:2510.05158*, 2025. URL <https://arxiv.org/abs/2510.05158>.
- [83] Zihong He and Changwang Zhang. AFSP: Agent Framework for Shaping Preference and Personality with Large Language Models. *arXiv preprint arXiv:2401.02870*, 2024. URL <https://arxiv.org/abs/2401.02870>.
- [84] Paul R. B. Houssel, Priyanka Singh, Siamak Layeghy, and Marius Portmann. Towards Explainable Network Intrusion Detection using Large Language Models. *arXiv preprint arXiv:2408.04342*, 2024. URL <https://arxiv.org/abs/2408.04342>.
- [85] Cheng-Yu Hsieh, Si-An Chen, Chun-Liang Li, Yasuhisa Fujii, Alexander Ratner, Chen-Yu Lee, Ranjay Krishna, and Tomas Pfister. Tool Documentation Enables Zero-Shot Tool-Usage with Large Lan-

- guage Models. *arXiv preprint arXiv:2308.00675*, 2023. URL <https://arxiv.org/abs/2308.00675>.
- [86] Jinchao Hu, Meizhi Zhong, Kehai Chen, Xuefeng Bai, and Min Zhang. Agentic Tool Use in Large Language Models. *arXiv preprint arXiv:2604.00835*, 2026. URL <https://arxiv.org/abs/2604.00835>.
- [87] Mengkang Hu, Pu Zhao, Can Xu, Qingfeng Sun, Jianguang Lou, Qingwei Lin, Ping Luo, and Saravan Rajmohan. AgentGen: Enhancing Planning Abilities for Large Language Model based Agent via Environment and Task Generation. *arXiv preprint arXiv:2408.00764*, 2024. URL <https://arxiv.org/abs/2408.00764>.
- [88] Yuncheng Hua, Lizhen Qu, and Gholamreza Haffari. Assistive Large Language Model Agents for Socially-Aware Negotiation Dialogues. *arXiv preprint arXiv:2402.01737*, 2024. URL <https://arxiv.org/abs/2402.01737>.
- [89] Shiting Huang, Zhen Fang, Zehui Chen, Siyu Yuan, Junjie Ye, Yu Zeng, Lin Chen, Qi Mao, and Feng Zhao. CRITICTOOL: Evaluating Self-Critique Capabilities of Large Language Models in Tool-Calling Error Scenarios. *arXiv preprint arXiv:2506.13977*, 2025. URL <https://arxiv.org/abs/2506.13977>.
- [90] Yue Huang, Jiawen Shi, Yuan Li, Chenrui Fan, Siyuan Wu, Qihui Zhang, Yixin Liu, Pan Zhou, Yao Wan, Neil Zhenqiang Gong, and Lichao Sun. MetaTool Benchmark for Large Language Models: Deciding Whether to Use Tools and Which to Use. *arXiv preprint arXiv:2310.03128*, 2023. URL <https://arxiv.org/abs/2310.03128>.
- [91] Daksh Jain, Aarya Jain, Ashutosh Desai, Avyakt Verma, Ishan Bhanuka, Pratik Narang, and Dhruv Kumar. Large Language Models as Pokemon Battle Agents: Strategic Play and Content Generation. *arXiv preprint arXiv:2512.17308*, 2025. URL <https://arxiv.org/abs/2512.17308>.
- [92] Kyochul Jang, Donghyeon Lee, Kyusik Kim, Dongseok Heo, Taewhoo Lee, Woojeong Kim, and Bongwon Suh. DICE-BENCH: Evaluating the Tool-Use Capabilities of Large Language Models in Multi-Round, Multi-Party Dialogues. *arXiv preprint arXiv:2506.22853*, 2025. URL <https://arxiv.org/abs/2506.22853>.
- [93] Adam Jelley, Yuhan Cao, Dave Bignell, Amos Storkey, Sam Devlin, and Tabish Rashid. Aligning Agents like Large Language Models. *arXiv preprint arXiv:2406.04208*, 2024. URL <https://arxiv.org/abs/2406.04208>.
- [94] Jingyi Jia and Qinbin Li. AutoTool: Efficient Tool Selection for Large Language Model Agents. *arXiv preprint arXiv:2511.14650*, 2025. URL <https://arxiv.org/abs/2511.14650>.
- [95] Feibo Jiang, Li Dong, Yubo Peng, Kezhi Wang, Kun Yang, Cunhua Pan, Dusit Niyato, and Octavia A. Dobre. Large Language Model Enhanced Multi-Agent Systems for 6G Communications. *arXiv preprint arXiv:2312.07850*, 2023. URL <https://arxiv.org/abs/2312.07850>.
- [96] Haitao Jiang, Lin Ge, Yuhe Gao, Jianian Wang, and Rui Song. LLM4Causal: Democratized Causal Tools for Everyone via Large Language Model. *arXiv preprint arXiv:2312.17122*, 2023. URL <https://arxiv.org/abs/2312.17122>.
- [97] Jinhao Jiang, Kun Zhou, Wayne Xin Zhao, Yang Song, Chen Zhu, Hengshu Zhu, and Ji-Rong Wen. KG-Agent: An Efficient Autonomous Agent Framework for Complex Reasoning over Knowledge Graph. *arXiv preprint arXiv:2402.11163*, 2024. URL <https://arxiv.org/abs/2402.11163>.
- [98] Cristian Jimenez-Romero, Alper Yegenoglu, and Christian Blum. Multi-Agent Systems Powered by Large Language Models: Applications in Swarm Intelligence. *arXiv preprint arXiv:2503.03800*, 2025. URL <https://arxiv.org/abs/2503.03800>.
- [99] Qiao Jin, Yifan Yang, Qingyu Chen, and Zhiyong Lu. GeneGPT: Augmenting Large Language Models with Domain Tools for Improved Access to Biomedical Information. *arXiv preprint arXiv:2304.09667*, 2023. URL <https://arxiv.org/abs/2304.09667>.
- [100] Vivek Kalyan and Martin Andrews. Reinforcement Learning for Long-Horizon Multi-Turn Search Agents. *arXiv preprint arXiv:2510.24126*, 2025. URL <https://arxiv.org/abs/2510.24126>.
- [101] Timotheus Kampik. Implementing Agents in JavaScript. *arXiv preprint arXiv:2505.18228*, 2025. URL <https://arxiv.org/abs/2505.18228>.
- [102] Xingjian Kang, Linda Vorberg, Andreas Maier, Alexander Katzmann, and Oliver Taubmann. Scando Attitude: Towards Autonomous CT Protocol Management using a Large Language Model Agent. *arXiv preprint arXiv:2509.20270*, 2025. URL <https://arxiv.org/abs/2509.20270>.
- [103] Kiran Kate, Tejaswini Pedapati, Kinjal Basu, Yara Rizk, Vijil Chenthamarakshan, Subhajit Chaudhury, Mayank Agarwal, and Ibrahim Abdelaziz. LongFuncEval: Measuring the effectiveness of long context models for function calling. *arXiv preprint arXiv:2505.10570*, 2025. URL <https://arxiv.org/abs/2505.10570>.
- [104] Ishan Kavathekar, Raghav Donakanti, Ponnu-rangam Kumaraguru, and Karthik Vaidhyanathan. Small Models, Big Tasks: An Exploratory Empirical Study on Small Language Models for Function Calling. *arXiv preprint arXiv:2504.19277*, 2025. URL <https://arxiv.org/abs/2504.19277>.

- [105] Jaekyeom Kim, Dong-Ki Kim, Lajanugen Logeswaran, Sungryull Sohn, and Honglak Lee. Auto-Intent: Automated Intent Discovery and Self-Exploration for Large Language Model Web Agents. *arXiv preprint arXiv:2410.22552*, 2024. URL <https://arxiv.org/abs/2410.22552>.
- [106] Juhyeong Kim, Yejin Kim, Youngbin Lee, and Hyunwoo Byun. FinAI Data Assistant: LLM-based Financial Database Query Processing with the OpenAI Function Calling API. *arXiv preprint arXiv:2510.14162*, 2025. URL <https://arxiv.org/abs/2510.14162>.
- [107] Lars Klein, Nearchos Potamitis, Roland Aydin, Robert West, Caglar Gulcehre, and Akhil Arora. Fleet of Agents: Coordinated Problem Solving with Large Language Models. *arXiv preprint arXiv:2405.06691*, 2024. URL <https://arxiv.org/abs/2405.06691>.
- [108] Naveen Krishnan. AI Agents: Evolution, Architecture, and Real-World Applications. *arXiv preprint arXiv:2503.12687*, 2025. URL <https://arxiv.org/abs/2503.12687>.
- [109] Zhiyi Kuang, Ryan Rong, YuCheng Yuan, and Allen Nie. Learning Game-Playing Agents with Generative Code Optimization. *arXiv preprint arXiv:2508.19506*, 2025. URL <https://arxiv.org/abs/2508.19506>.
- [110] Younes Lakhnati, Max Pascher, and Jens Gerken. Exploring Large Language Models to Facilitate Variable Autonomy for Human-Robot Teaming. *arXiv preprint arXiv:2312.07214*, 2023. URL <https://arxiv.org/abs/2312.07214>.
- [111] Payam Latifi. Is 'Hope' a person or an idea? A pilot benchmark for NER: comparing traditional NLP tools and large language models on ambiguous entities. *arXiv preprint arXiv:2509.12098*, 2025. URL <https://arxiv.org/abs/2509.12098>.
- [112] Nancy Lau, Louis Sloot, Jyoutir Raj, Giuseppe Marco Boscardin, Evan Harris, Dylan Bowman, Mario Brajkovski, Jaideep Chawla, and Dan Zhao. ZeroDayBench: Evaluating LLM Agents on Unseen Zero-Day Vulnerabilities for Cyberdefense. *arXiv preprint arXiv:2603.02297*, 2026. URL <https://arxiv.org/abs/2603.02297>.
- [113] Dong Won Lee, Hae Won Park, Cynthia Breazeal, and Louis-Philippe Morency. Aligning Dialogue Agents with Global Feedback via Large Language Model Multimodal Reward Decomposition. *arXiv preprint arXiv:2505.15922*, 2025. URL <https://arxiv.org/abs/2505.15922>.
- [114] Chenghao Li, Liu Liu, Baosheng Yu, Jiayan Qiu, and Yibing Zhan. Re-Initialization Token Learning for Tool-Augmented Large Language Models. *arXiv preprint arXiv:2506.14248*, 2025. URL <https://arxiv.org/abs/2506.14248>.
- [115] Dawei Li, Zhen Tan, Peijia Qian, Yifan Li, Kumar Satvik Chaudhary, Lijie Hu, and Jiayi Shen. SMoA: Improving Multi-agent Large Language Models with Sparse Mixture-of-Agents. *arXiv preprint arXiv:2411.03284*, 2024. URL <https://arxiv.org/abs/2411.03284>.
- [116] Huao Li, Yu Quan Chong, Simon Stepputtis, Joseph Campbell, Dana Hughes, Michael Lewis, and Katia Sycara. Theory of Mind for Multi-Agent Collaboration via Large Language Models. *arXiv preprint arXiv:2310.10701*, 2023. URL <https://arxiv.org/abs/2310.10701>.
- [117] Ning Li, Xiangmou Qu, Jiamu Zhou, Jun Wang, Muning Wen, Kounianhua Du, Xingyu Lou, Qiuying Peng, Jun Wang, and Weinan Zhang. MobileUse: A GUI Agent with Hierarchical Reflection for Autonomous Mobile Operation. *arXiv preprint arXiv:2507.16853*, 2025. URL <https://arxiv.org/abs/2507.16853>.
- [118] Wenjun Li, Dexun Li, Kuicai Dong, Cong Zhang, Hao Zhang, Weiwen Liu, Yasheng Wang, Ruiming Tang, and Yong Liu. Adaptive Tool Use in Large Language Models with Meta-Cognition Trigger. *arXiv preprint arXiv:2502.12961*, 2025. URL <https://arxiv.org/abs/2502.12961>.
- [119] Xingzuo Li, Kehai Chen, Yunfei Long, Xuefeng Bai, Yong Xu, and Min Zhang. Generator-Assistant Stepwise Rollback Framework for Large Language Model Agent. *arXiv preprint arXiv:2503.02519*, 2025. URL <https://arxiv.org/abs/2503.02519>.
- [120] Yongxiang Li, Moxin Li, Zhixin Ma, Fengbin Zhu, Dongrui Liu, Wenjie Wang, and Fuli Feng. Plant, Persist, Trigger: Sleeper Attack on Large Language Model Agents. *arXiv preprint arXiv:2605.28201*, 2026. URL <https://arxiv.org/abs/2605.28201>.
- [121] Zelong Li, Shuyuan Xu, Kai Mei, Wenyue Hua, Balaji Rama, Om Raheja, Hao Wang, He Zhu, and Yongfeng Zhang. AutoFlow: Automated Workflow Generation for Large Language Model Agents. *arXiv preprint arXiv:2407.12821*, 2024. URL <https://arxiv.org/abs/2407.12821>.
- [122] Zeping Li, Hongru Wang, Yiwen Zhao, Guanhua Chen, Yixia Li, Keyang Chen, Yixin Cao, Guangnan Ye, Hongfeng Chai, and Zhenfei Yin. Rethinking the Role of Entropy in Optimizing Tool-Use Behaviors for Large Language Model Agents. *arXiv preprint arXiv:2602.02050*, 2026. URL <https://arxiv.org/abs/2602.02050>.
- [123] Guoqiang Liang, Jingqian Gong, Mengxuan Li, Gege Lin, and Shuo Zhang. The Empowerment of Science of Science by Large Language Models: New Tools and Methods. *arXiv preprint*

- arXiv:2511.15370*, 2025. URL <https://arxiv.org/abs/2511.15370>.
- [124] Haoran Liang, Yufa Zhou, Mohammad Talebi Kalaleh, and Qipei Mei. Automating Structural Engineering Workflows with Large Language Model Agents. *arXiv preprint arXiv:2510.11004*, 2025. URL <https://arxiv.org/abs/2510.11004>.
- [125] Jonghan Lim, Birgit Vogel-Heuser, and Ilya Kovalenko. Large Language Model-Enabled Multi-Agent Manufacturing Systems. *arXiv preprint arXiv:2406.01893*, 2024. URL <https://arxiv.org/abs/2406.01893>.
- [126] Guang Lin, Toshihisa Tanaka, and Qibin Zhao. Large Language Model Sentinel: LLM Agent for Adversarial Purification. *arXiv preprint arXiv:2405.20770*, 2024. URL <https://arxiv.org/abs/2405.20770>.
- [127] Jianghao Lin, Yuanyuan Shi, Xin Peng, Renjie Ding, Hairui Wang, Yuxuan Peng, Bizhe Bai, Weixi Song, Fengshuo Bai, Huacan Chai, Weinan Zhang, Fei Huang, and Ying Wen. Tool-PRM: Fine-Grained Inference Scaling of Structured Outputs for Function Calling. *arXiv preprint arXiv:2510.14703*, 2025. URL <https://arxiv.org/abs/2510.14703>.
- [128] Jianghao Lin, Xinyuan Wang, Xinyi Dai, Menghui Zhu, Bo Chen, Ruiming Tang, Yong Yu, and Weinan Zhang. MassTool: A Multi-Task Search-Based Tool Retrieval Framework for Large Language Models. *arXiv preprint arXiv:2507.00487*, 2025. URL <https://arxiv.org/abs/2507.00487>.
- [129] Qingming Lin, Rui Hu, Huaxia Li, Sensen Wu, Yadong Li, Kai Fang, Hailin Feng, Zhenhong Du, and Liuchang Xu. ShapefileGPT: A Multi-Agent Large Language Model Framework for Automated Shapefile Processing. *arXiv preprint arXiv:2410.12376*, 2024. URL <https://arxiv.org/abs/2410.12376>.
- [130] Qiqiang Lin, Muning Wen, Qiuying Peng, Guanyu Nie, Junwei Liao, Jun Wang, Xiaoyun Mo, Jiamu Zhou, Cheng Cheng, Yin Zhao, Jun Wang, and Weinan Zhang. Hammer: Robust Function-Calling for On-Device Language Models via Function Masking. *arXiv preprint arXiv:2410.04587*, 2024. URL <https://arxiv.org/abs/2410.04587>.
- [131] Zihan Lin, Xiaohan Wang, Jie Cao, Jiajun Chai, Guojun Yin, Wei Lin, and Ran He. ResT: Reshaping Token-Level Policy Gradients for Tool-Use Large Language Models. *arXiv preprint arXiv:2509.21826*, 2025. URL <https://arxiv.org/abs/2509.21826>.
- [132] Changshu Liu and Reyhaneh Jabbarvand. A Tool for In-depth Analysis of Code Execution Reasoning of Large Language Models. *arXiv preprint arXiv:2501.18482*, 2025. URL <https://arxiv.org/abs/2501.18482>.
- [133] Hanbing Liu, Chunhao Tian, Nan An, Ziyuan Wang, Pinyan Lu, Changyuan Yu, and Qi Qi. Budget-Constrained Agentic Large Language Models: Intention-Based Planning for Costly Tool Use. *arXiv preprint arXiv:2602.11541*, 2026. URL <https://arxiv.org/abs/2602.11541>.
- [134] Jun Liu, Zhenglun Kong, Peiyan Dong, Changdi Yang, Tianqi Li, Hao Tang, Geng Yuan, Wei Niu, Wenbin Zhang, Pu Zhao, Xue Lin, Dong Huang, and Yanzhi Wang. Structured Agent Distillation for Large Language Model. *arXiv preprint arXiv:2505.13820*, 2025. URL <https://arxiv.org/abs/2505.13820>.
- [135] Siyu Liu, Bo Hu, Beilin Ye, Jiamin Xu, David J. Srolovitz, and Tongqi Wen. MatTools: Benchmarking Large Language Models for Materials Science Tools. *arXiv preprint arXiv:2505.10852*, 2025. URL <https://arxiv.org/abs/2505.10852>.
- [136] Weiwen Liu, Xu Huang, Xingshan Zeng, Xinlong Hao, Shuai Yu, Dexun Li, Shuai Wang, Weinan Gan, Zhengying Liu, Yuanqing Yu, Zezhong Wang, Yuxian Wang, Wu Ning, Yutai Hou, Bin Wang, Chuhan Wu, Xinzhi Wang, Yong Liu, Yasheng Wang, Duyu Tang, et al. ToolACE: Winning the Points of LLM Function Calling. *arXiv preprint arXiv:2409.00920*, 2024. URL <https://arxiv.org/abs/2409.00920>.
- [137] Xukun Liu, Zhiyuan Peng, Xiaoyuan Yi, Xing Xie, Lirong Xiang, Yuchen Liu, and Dongkuan Xu. ToolNet: Connecting Large Language Models with Massive Tools via Tool Graph. *arXiv preprint arXiv:2403.00839*, 2024. URL <https://arxiv.org/abs/2403.00839>.
- [138] Yang Liu, Peng Sun, and Hang Li. Large Language Models as Agents in Two-Player Games. *arXiv preprint arXiv:2402.08078*, 2024. URL <https://arxiv.org/abs/2402.08078>.
- [139] Yifei Liu, Yu Cui, and Haibin Zhang. RRTL: Red Teaming Reasoning Large Language Models in Tool Learning. *arXiv preprint arXiv:2505.17106*, 2025. URL <https://arxiv.org/abs/2505.17106>.
- [140] Yuyang Liu, Meng Cao, Xinyuan Shi, and Xiaondan Liang. COLT: Enhancing Video Large Language Models with Continual Tool Usage. *arXiv preprint arXiv:2509.18754*, 2025. URL <https://arxiv.org/abs/2509.18754>.
- [141] Zefang Liu. Multi-Agent Collaboration in Incident Response with Large Language Models. *arXiv preprint arXiv:2412.00652*, 2024. URL <https://arxiv.org/abs/2412.00652>.
- [142] Ziqi. Liu, Ziyang. Zhou, and Mingxuan. Hu. CAF-I: A Collaborative Multi-Agent Framework for En-

- hanced Irony Detection with Large Language Models. *arXiv preprint arXiv:2506.08430*, 2025. URL <https://arxiv.org/abs/2506.08430>.
- [143] Joseph R. Loffredo and Suyeol Yun. Agent-Enhanced Large Language Models for Researching Political Institutions. *arXiv preprint arXiv:2503.13524*, 2025. URL <https://arxiv.org/abs/2503.13524>.
- [144] Alejandro Lopez-Lira. Can Large Language Models Trade? Testing Financial Theories with LLM Agents in Market Simulations. *arXiv preprint arXiv:2504.10789*, 2025. URL <https://arxiv.org/abs/2504.10789>.
- [145] Junyu Luo, Weizhi Zhang, Ye Yuan, Yusheng Zhao, Junwei Yang, Yiyang Gu, Bohan Wu, Binqi Chen, Ziyue Qiao, Qingqing Long, Rongcheng Tu, Xiao Luo, Wei Ju, Zhiping Xiao, Yifan Wang, Meng Xiao, Chenwu Liu, Jingyang Yuan, Shichang Zhang, Yiqiao Jin, et al. Large Language Model Agent: A Survey on Methodology, Applications and Challenges. *arXiv preprint arXiv:2503.21460*, 2025. URL <https://arxiv.org/abs/2503.21460>.
- [146] Ne Luo, Aryo Pradipta Gema, Xuanli He, Emile van Krieken, Pietro Lesci, and Pasquale Minervini. Self-Training Large Language Models for Tool-Use Without Demonstrations. *arXiv preprint arXiv:2502.05867*, 2025. URL <https://arxiv.org/abs/2502.05867>.
- [147] Qianqian Luo, Qingming Lin, Liuchang Xu, Sensen Wu, Ruichen Mao, Chao Wang, Hailin Feng, Bo Huang, and Zhenhong Du. GeoJSON Agents: A Multi-Agent LLM Architecture for Geospatial Analysis-Function Calling vs Code Generation. *arXiv preprint arXiv:2509.08863*, 2025. URL <https://arxiv.org/abs/2509.08863>.
- [148] Zheng Luo, T Pranav Kutralingam, Ogochukwu N Okoani, Wanpeng Xu, Hua Wei, and Xiyang Hu. Lost in Execution: On the Multilingual Robustness of Tool Calling in Large Language Models. *arXiv preprint arXiv:2601.05366*, 2026. URL <https://arxiv.org/abs/2601.05366>.
- [149] Heng Ma, Alexander Brace, Carlo Siebenschuh, Greg Pauloski, Ian Foster, and Arvind Ramanathan. Connecting Large Language Model Agent to High Performance Computing Resource. *arXiv preprint arXiv:2502.12280*, 2025. URL <https://arxiv.org/abs/2502.12280>.
- [150] Zhiyuan Ma, Jiayu Liu, Xianzhen Luo, Zhenya Huang, Qingfu Zhu, and Wanxiang Che. Advancing Tool-Augmented Large Language Models via Meta-Verification and Reflection Learning. *arXiv preprint arXiv:2506.04625*, 2025. URL <https://arxiv.org/abs/2506.04625>.
- [151] Chris Madge and Massimo Poesio. Large Language Models as Minecraft Agents. *arXiv preprint arXiv:2402.08392*, 2024. URL <https://arxiv.org/abs/2402.08392>.
- [152] Bodhisattwa Prasad Majumder, Harshit Surana, Dhruv Agarwal, Bhavana Dalvi Mishra, Abhijeetsingh Meena, Aryan Prakhar, Tirth Vora, Tushar Khot, Ashish Sabharwal, and Peter Clark. DiscoveryBench: Towards Data-Driven Discovery with Large Language Models. *arXiv preprint arXiv:2407.01725*, 2024. URL <https://arxiv.org/abs/2407.01725>.
- [153] Stephanie Anneris Malvicini, Ewelina Gajewska, Arda Derbent, Katarzyna Budzynska, Jarosaw A. Chudziak, and Maria Vanina Martinez. A Natural Language Agentic Approach to Study Affective Polarization. *arXiv preprint arXiv:2603.02711*, 2026. URL <https://arxiv.org/abs/2603.02711>.
- [154] Graziano A. Manduzio, Federico A. Galatolo, Mario G. C. A. Cimino, Enzo Pasquale Scilingo, and Lorenzo Cominelli. Improving Small-Scale Large Language Models Function Calling for Reasoning Tasks. *arXiv preprint arXiv:2410.18890*, 2024. URL <https://arxiv.org/abs/2410.18890>.
- [155] Prattyush Mangal, Carol Mak, Theo Kanakis, Timothy Donovan, Dave Braines, and Edward Pyzer-Knapp. Coalitions of Large Language Models Increase the Robustness of AI Agents. *arXiv preprint arXiv:2408.01380*, 2024. URL <https://arxiv.org/abs/2408.01380>.
- [156] Jiageng Mao, Junjie Ye, Yuxi Qian, Marco Pavone, and Yue Wang. A Language Agent for Autonomous Driving. *arXiv preprint arXiv:2311.10813*, 2023. URL <https://arxiv.org/abs/2311.10813>.
- [157] Samuele Marro, Alan Chan, Xinxing Ren, Lewis Hammond, Jesse Wright, Gurjyot Wanga, Tiziano Piccardi, Nuno Campos, Tobin South, Jialin Yu, Sunando Sengupta, Eric Sommerlade, Alex Pentland, Philip Torr, and Jiaxin Pei. Permission Manifests for Web Agents. *arXiv preprint arXiv:2601.02371*, 2025. URL <https://arxiv.org/abs/2601.02371>.
- [158] Zohaib Masood and Miguel Vargas Martin. Beyond Static Tools: Evaluating Large Language Models for Cryptographic Misuse Detection. *arXiv preprint arXiv:2411.09772*, 2024. URL <https://arxiv.org/abs/2411.09772>.
- [159] Charlie Masters, Marta Grzeskiewicz, and Stefano V. Albrecht. ARCANE: A Multi-Agent Framework for Interpretable and Configurable Alignment. *arXiv preprint arXiv:2512.06196*, 2025. URL <https://arxiv.org/abs/2512.06196>.
- [160] Gabriele Mattioli, Evelyn Turri, Sara Sarto, Lorenzo Baraldi, Marcella Cornia, Lorenzo Baraldi, and Rita Cucchiara. RaTA-Tool: Retrieval-based Tool Selection with Multimodal Large Language

- Models. *arXiv preprint arXiv:2604.14951*, 2026. URL <https://arxiv.org/abs/2604.14951>.
- [161] Mike A. Merrill, Akshay Paruchuri, Naghmeh Rezaei, Geza Kovacs, Javier Perez, Yun Liu, Erik Schenck, Nova Hammerquist, Jake Sunshine, Shyam Tailor, Kumar Ayush, Hao-Wei Su, Qian He, Cory Y. McLean, Mark Malhotra, Shwetak Patel, Jiening Zhan, Tim Althoff, Daniel McDuff, and Xin Liu. Transforming Wearable Data into Personal Health Insights using Large Language Model Agents. *arXiv preprint arXiv:2406.06464*, 2024. URL <https://arxiv.org/abs/2406.06464>.
- [162] Mohammad Mahdi Mohajer, Reem Aleithan, Nima Shiri Harzevili, Moshi Wei, Alvine Boaye Belle, Hung Viet Pham, and Song Wang. Skip-Analyzer: A Tool for Static Code Analysis with Large Language Models. *arXiv preprint arXiv:2310.18532*, 2023. URL <https://arxiv.org/abs/2310.18532>.
- [163] Sarat Mudunuri, Jian Wan, Ally Qin, and Srinivasan Manoharan. Semantic Tool Discovery for Large Language Models: A Vector-Based Approach to MCP Tool Selection. *arXiv preprint arXiv:2603.20313*, 2026. URL <https://arxiv.org/abs/2603.20313>.
- [164] Jiliang Ni, Jiachen Pu, Zhongyi Yang, Jingfeng Luo, and Conggang Hu. STAR: Similarity-guided Teacher-Assisted Refinement for Super-Tiny Function Calling Models. *arXiv preprint arXiv:2602.03022*, 2026. URL <https://arxiv.org/abs/2602.03022>.
- [165] Shiwen Ni, Guhong Chen, Shuaimin Li, Xuanang Chen, Siyi Li, Bingli Wang, Qiyao Wang, Xingjian Wang, Yifan Zhang, Liyang Fan, Chengming Li, Ruifeng Xu, Le Sun, and Min Yang. A Survey on Large Language Model Benchmarks. *arXiv preprint arXiv:2508.15361*, 2025. URL <https://arxiv.org/abs/2508.15361>.
- [166] Xiaohang Nie, Zihan Guo, Youliang Chen, Yuanjian Zhou, and Weinan Zhang. AWCP: A Workspace Delegation Protocol for Deep-Engagement Collaboration across Remote Agents. *arXiv preprint arXiv:2602.20493*, 2026. URL <https://arxiv.org/abs/2602.20493>.
- [167] Nripesh Niketan and Hadj Batatia. Integrating External Tools with Large Language Models to Improve Accuracy. *arXiv preprint arXiv:2507.08034*, 2025. URL <https://arxiv.org/abs/2507.08034>.
- [168] Kangyun Ning, Yisong Su, Xueqiang Lv, Yuanzhe Zhang, Jian Liu, Kang Liu, and Jinan Xu. WTU-EVAL: A Whether-or-Not Tool Usage Evaluation Benchmark for Large Language Models. *arXiv preprint arXiv:2407.12823*, 2024. URL <https://arxiv.org/abs/2407.12823>.
- [169] Xuying Ning, Katherine Tieu, Dongqi Fu, Tianxin Wei, Zihao Li, Yuanchen Bei, Jiaru Zou, Mengting Ai, Zhining Liu, Ting-Wei Li, Lingjie Chen, Yanjun Zhao, Ke Yang, Bingxuan Li, Cheng Qian, Gaotang Li, Xiao Lin, Zhichen Zeng, Ruizhong Qiu, Sirui Chen, et al. Code as Agent Harness. *arXiv preprint arXiv:2605.18747*, 2026. URL <https://arxiv.org/abs/2605.18747>.
- [170] Janghoon Ock, Radheesh Sharma Meda, Tirtha Vinchurkar, Yayati Jadhav, and Amir Barati Farimani. Adsorb-Agent: Autonomous Identification of Stable Adsorption Configurations via Large Language Model Agent. *arXiv preprint arXiv:2410.16658*, 2024. URL <https://arxiv.org/abs/2410.16658>.
- [171] Kunal Pai, Parth Shah, and Harshil Patel. NAAMSE: Framework for Evolutionary Security Evaluation of Agents. *arXiv preprint arXiv:2602.07391*, 2026. URL <https://arxiv.org/abs/2602.07391>.
- [172] Norman Di Palo, Leonard Hasenclever, Jan Humpalik, and Arunkumar Byravan. Diffusion Augmented Agents: A Framework for Efficient Exploration and Transfer Learning. *arXiv preprint arXiv:2407.20798*, 2024. URL <https://arxiv.org/abs/2407.20798>.
- [173] Lihang Pan, Yuxuan Li, Chun Yu, and Yuanchun Shi. A Human-Computer Collaborative Tool for Training a Single Large Language Model Agent into a Network through Few Examples. *arXiv preprint arXiv:2404.15974*, 2024. URL <https://arxiv.org/abs/2404.15974>.
- [174] Jing-Cheng Pang, Si-Hang Yang, Kaiyuan Li, Jiaji Zhang, Xiong-Hui Chen, Nan Tang, and Yang Yu. Knowledgeable Agents by Offline Reinforcement Learning from Large Language Model Rollouts. *arXiv preprint arXiv:2404.09248*, 2024. URL <https://arxiv.org/abs/2404.09248>.
- [175] Varatheepan Paramanayakam, Andreas Karatzas, Iraklis Anagnostopoulos, and Dimitrios Stamoulis. Less is More: Optimizing Function Calling for LLM Execution on Edge Devices. *arXiv preprint arXiv:2411.15399*, 2024. URL <https://arxiv.org/abs/2411.15399>.
- [176] Varatheepan Paramanayakam, Andreas Karatzas, Iraklis Anagnostopoulos, and Dimitrios Stamoulis. CarbonCall: Sustainability-Aware Function Calling for Large Language Models on Edge Devices. *arXiv preprint arXiv:2504.20348*, 2025. URL <https://arxiv.org/abs/2504.20348>.
- [177] Bhargavi Paranjape, Scott Lundberg, Sameer Singh, Hannaneh Hajishirzi, Luke Zettlemoyer, and Marco Tulio Ribeiro. ART: Automatic multi-step reasoning and tool-use for large language mod-

- els. *arXiv preprint arXiv:2303.09014*, 2023. URL <https://arxiv.org/abs/2303.09014>.
- [178] Ajay Patel, Markus Hofmarcher, Claudiu Leoveanu-Condrei, Marius-Constantin Dinu, Chris Callison-Burch, and Sepp Hochreiter. Large Language Models Can Self-Improve At Web Agent Tasks. *arXiv preprint arXiv:2405.20309*, 2024. URL <https://arxiv.org/abs/2405.20309>.
- [179] Bhrij Patel, Davide Belli, Amir Jalalirad, Maximilian Arnold, Aleksandr Ermolov, and Bence Major. Dynamic Tool Dependency Retrieval for Lightweight Function Calling. *arXiv preprint arXiv:2512.17052*, 2025. URL <https://arxiv.org/abs/2512.17052>.
- [180] Zhiyuan Peng, Xin Yin, Rui Qian, Peiqin Lin, Yongkang Liu, Hao Zhang, Chenhao Ying, and Yuan Luo. SolEval: Benchmarking Large Language Models for Repository-level Solidity Code Generation. *arXiv preprint arXiv:2502.18793*, 2025. URL <https://arxiv.org/abs/2502.18793>.
- [181] Pouya Pezeshkpour and Estevam Hruschka. AutoPyVerifier: Learning Compact Executable Verifiers for Large Language Model Outputs. *arXiv preprint arXiv:2604.22937*, 2026. URL <https://arxiv.org/abs/2604.22937>.
- [182] Aske Plaat, Max van Duijn, Niki van Stein, Mike Preuss, Peter van der Putten, and Kees Joost Batenburg. Agentic Large Language Models, a survey. *arXiv preprint arXiv:2503.23037*, 2025. URL <https://arxiv.org/abs/2503.23037>.
- [183] Ji Qi, Kaixuan Ji, Jifan Yu, Duokang Wang, Bin Xu, Lei Hou, and Juanzi Li. VidCoM: Fast Video Comprehension through Large Language Models with Multimodal Tools. *arXiv preprint arXiv:2310.10586*, 2023. URL <https://arxiv.org/abs/2310.10586>.
- [184] Yunjia Qi, Hao Peng, Xiaozhi Wang, Amy Xin, Youfeng Liu, Bin Xu, Lei Hou, and Juanzi Li. AGENTIF: Benchmarking Instruction Following of Large Language Models in Agentic Scenarios. *arXiv preprint arXiv:2505.16944*, 2025. URL <https://arxiv.org/abs/2505.16944>.
- [185] Cheng Qian, Chi Han, Yi R. Fung, Yujia Qin, Zhiyuan Liu, and Heng Ji. CREATOR: Tool Creation for Disentangling Abstract and Concrete Reasoning of Large Language Models. *arXiv preprint arXiv:2305.14318*, 2023. URL <https://arxiv.org/abs/2305.14318>.
- [186] Xihe Qiu, Haoyu Wang, Xiaoyu Tan, Chao Qu, Yujie Xiong, Yuan Cheng, Yinghui Xu, Wei Chu, and Yuan Qi. Towards Collaborative Intelligence: Propagating Intentions and Reasoning for Multi-Agent Coordination with Large Language Models. *arXiv preprint arXiv:2407.12532*, 2024. URL <https://arxiv.org/abs/2407.12532>.
- [187] Changle Qu, Sunhao Dai, Xiaochi Wei, Hengyi Cai, Shuaiqiang Wang, Dawei Yin, Jun Xu, and Ji-Rong Wen. Tool Learning with Large Language Models: A Survey. *arXiv preprint arXiv:2405.17935*, 2024. URL <https://arxiv.org/abs/2405.17935>.
- [188] Changle Qu, Sunhao Dai, Xiaochi Wei, Hengyi Cai, Shuaiqiang Wang, Dawei Yin, Jun Xu, and Ji-Rong Wen. Towards Completeness-Oriented Tool Retrieval for Large Language Models. *arXiv preprint arXiv:2405.16089*, 2024. URL <https://arxiv.org/abs/2405.16089>.
- [189] Ella Rabinovich and Ateret Anaby-Tavor. On the Robustness of Agentic Function Calling. *arXiv preprint arXiv:2504.00914*, 2025. URL <https://arxiv.org/abs/2504.00914>.
- [190] Nate Rahn, Pierluca D’Oro, and Marc G. Bellemare. Controlling Large Language Model Agents with Entropic Activation Steering. *arXiv preprint arXiv:2406.00244*, 2024. URL <https://arxiv.org/abs/2406.00244>.
- [191] Hossein Rajabzadeh, Suyuchen Wang, Hyock Ju Kwon, and Bang Liu. Multimodal Multi-Hop Question Answering Through a Conversation Between Tools and Efficiently Finetuned Large Language Models. *arXiv preprint arXiv:2309.08922*, 2023. URL <https://arxiv.org/abs/2309.08922>.
- [192] Mayk Caldas Ramos, Christopher J. Collison, and Andrew D. White. A Review of Large Language Models and Autonomous Agents in Chemistry. *arXiv preprint arXiv:2407.01603*, 2024. URL <https://arxiv.org/abs/2407.01603>.
- [193] Swanand Rao. Tool Forge: A Validation-Carrying Toolchain for Governed Agentic Execution. *arXiv preprint arXiv:2605.28000*, 2026. URL <https://arxiv.org/abs/2605.28000>.
- [194] Swanand Rao, Kiran Kashalkar, Parvathi Somashekar, and Priya Krishnan. Aethon: A Reference-Based Replication Primitive for Constant-Time Instantiation of Stateful AI Agents. *arXiv preprint arXiv:2604.12129*, 2026. URL <https://arxiv.org/abs/2604.12129>.
- [195] Zeeshan Rasheed, Muhammad Waseem, Kari Systs, and Pekka Abrahamsson. Large Language Model Evaluation Via Multi AI Agents: Preliminary results. *arXiv preprint arXiv:2404.01023*, 2024. URL <https://arxiv.org/abs/2404.01023>.
- [196] Matthew Renze and Erhan Guven. Self-Reflection in LLM Agents: Effects on Problem-Solving Performance. *arXiv preprint arXiv:2405.06682*, 2024. URL <https://arxiv.org/abs/2405.06682>.
- [197] Fabio Cortes Rodriguez and Luciano Abriata. Speech recognition assisted by large language models to command software orally – Application to

- an augmented and virtual reality web app for immersive molecular graphics. *arXiv preprint arXiv:2603.02901*, 2026. URL <https://arxiv.org/abs/2603.02901>.
- [198] Samuel Rothfarb, Megan C. Davis, Ivana Matanovic, Baikun Li, Edward F. Holby, and Wilton J. M. Kort-Kamp. Hierarchical Multi-agent Large Language Model Reasoning for Autonomous Functional Materials Discovery. *arXiv preprint arXiv:2512.13930*, 2025. URL <https://arxiv.org/abs/2512.13930>.
- [199] Yuki Sakamoto, Takahisa Uchida, and Hiroshi Ishiguro. Value-Based Large Language Model Agent Simulation for Mutual Evaluation of Trust and Interpersonal Closeness. *arXiv preprint arXiv:2507.11979*, 2025. URL <https://arxiv.org/abs/2507.11979>.
- [200] Gabriel Sarch, Yue Wu, Michael J. Tarr, and Katerina Fragkiadaki. Open-Ended Instructable Embodied Agents with Memory-Augmented Large Language Models. *arXiv preprint arXiv:2310.15127*, 2023. URL <https://arxiv.org/abs/2310.15127>.
- [201] Gabriel Sarch, Sahil Somani, Raghav Kapoor, Michael J. Tarr, and Katerina Fragkiadaki. HELPER-X: A Unified Instructable Embodied Agent to Tackle Four Interactive Vision-Language Domains with Memory-Augmented Language Models. *arXiv preprint arXiv:2404.19065*, 2024. URL <https://arxiv.org/abs/2404.19065>.
- [202] Fozle Rabbi Shafi, M. Anwar Hossain, and Salimur Choudhury. mind_call: A Dataset for Mental Health Function Calling with Large Language Models. *arXiv preprint arXiv:2601.06937*, 2026. URL <https://arxiv.org/abs/2601.06937>.
- [203] Yunfan Shao, Linyang Li, Junqi Dai, and Xipeng Qiu. Character-LLM: A Trainable Agent for Role-Playing. *arXiv preprint arXiv:2310.10158*, 2023. URL <https://arxiv.org/abs/2310.10158>.
- [204] Sarath Shekizhar, Romain Cosentino, Adam Earle, and Silvio Savarese. Echoing: Identity Failures when LLM Agents Talk to Each Other. *arXiv preprint arXiv:2511.09710*, 2025. URL <https://arxiv.org/abs/2511.09710>.
- [205] Kou Shi, Ziao Zhang, Shiting Huang, Avery Nie, Zhen Fang, Qiuchen Wang, Lin Chen, Huaian Chen, Zehui Chen, and Feng Zhao. AsyncTool: Evaluating the Asynchronous Function Calling Capability under Multi-Task Scenarios. *arXiv preprint arXiv:2605.27995*, 2026. URL <https://arxiv.org/abs/2605.27995>.
- [206] Zhengliang Shi, Yuhan Wang, Lingyong Yan, Pengjie Ren, Shuaiqiang Wang, Dawei Yin, and Zhaochun Ren. Retrieval Models Aren't Tool-Savvy: Benchmarking Tool Retrieval for Large Language Models. *arXiv preprint arXiv:2503.01763*, 2025. URL <https://arxiv.org/abs/2503.01763>.
- [207] Connor Shorten, Charles Pierse, Thomas Benjamin Smith, Karel D'Oosterlinck, Tuana Celik, Erika Cardenas, Leonie Monigatti, Mohd Shukri Hasan, Edward Schmuhl, Daniel Williams, Aravind Kesiraju, and Bob van Luijt. Querying Databases with Function Calling. *arXiv preprint arXiv:2502.00032*, 2025. URL <https://arxiv.org/abs/2502.00032>.
- [208] Raphael Shu, Nilaksh Das, Michelle Yuan, Monica Sunkara, and Yi Zhang. Towards Effective GenAI Multi-Agent Collaboration: Design and Evaluation for Enterprise Applications. *arXiv preprint arXiv:2412.05449*, 2024. URL <https://arxiv.org/abs/2412.05449>.
- [209] Simranjit Singh, Andreas Karatzas, Michael Fore, Iraklis Anagnostopoulos, and Dimitrios Stamoulis. An LLM-Tool Compiler for Fused Parallel Function Calling. *arXiv preprint arXiv:2405.17438*, 2024. URL <https://arxiv.org/abs/2405.17438>.
- [210] Nikolai Skripko. Instruction-Following Evaluation in Function Calling for Large Language Models. *arXiv preprint arXiv:2509.18420*, 2025. URL <https://arxiv.org/abs/2509.18420>.
- [211] Chan Hee Song, Jiaman Wu, Clayton Washington, Brian M. Sadler, Wei-Lun Chao, and Yu Su. LLM-Planner: Few-Shot Grounded Planning for Embodied Agents with Large Language Models. *arXiv preprint arXiv:2212.04088*, 2022. URL <https://arxiv.org/abs/2212.04088>.
- [212] Yewei Song, Xunzhu Tang, Cedric Lothritz, Saad Ezzini, Jacques Klein, Tegawende F. Bissyande, Andrey Boytsov, Ulrick Ble, and Anne Goujon. CallNavi, A Challenge and Empirical Study on LLM Function Calling and Routing. *arXiv preprint arXiv:2501.05255*, 2025. URL <https://arxiv.org/abs/2501.05255>.
- [213] Asher Sprigler, Alexander Drobek, Keagan Weinstock, Wendpanga Tapsoba, Gavin Childress, Andy Dao, and Lucas Gral. Synergistic Simulations: Multi-Agent Problem Solving with Large Language Models. *arXiv preprint arXiv:2409.13753*, 2024. URL <https://arxiv.org/abs/2409.13753>.
- [214] Theodore R. Sumers, Shunyu Yao, Karthik Narasimhan, and Thomas L. Griffiths. Cognitive Architectures for Language Agents. *arXiv preprint arXiv:2309.02427*, 2023. URL <https://arxiv.org/abs/2309.02427>.
- [215] Haochen Sun, Shuwen Zhang, Lujie Niu, Lei Ren, Hao Xu, Hao Fu, Fangkun Zhao, Caixia Yuan, and Xiaojie Wang. Collab-Overcooked: Benchmarking

- and Evaluating Large Language Models as Collaborative Agents. *arXiv preprint arXiv:2502.20073*, 2025. URL <https://arxiv.org/abs/2502.20073>.
- [216] Maojun Sun, Ruijian Han, Binyan Jiang, Houduo Qi, Defeng Sun, Yancheng Yuan, and Jian Huang. A Survey on Large Language Model-based Agents for Statistics and Data Science. *arXiv preprint arXiv:2412.14222*, 2024. URL <https://arxiv.org/abs/2412.14222>.
- [217] Kamil Szczepanik and Jarosław A. Chudziak. TRIZ Agents: A Multi-Agent LLM Approach for TRIZ-Based Innovation. *arXiv preprint arXiv:2506.18783*, 2025. URL <https://arxiv.org/abs/2506.18783>.
- [218] Xing Tang, Hao Chen, Shiwei Li, Fuyuan Lyu, Weijie Shi, Lingjie Li, Dugang Liu, Weihong Luo, Xiku Du, and Xiuqiang He. Data-Driven Function Calling Improvements in Large Language Model for Online Financial QA. *arXiv preprint arXiv:2604.05387*, 2026. URL <https://arxiv.org/abs/2604.05387>.
- [219] AMAP AI Agent Team, Yulan Hu, Xiangwen Zhang, Sheng Ouyang, Hao Yi, Lu Xu, Qinglin Lang, Lide Tan, Xiang Cheng, Tianchen Ye, Zhicong Li, Ge Chen, Wenjin Yang, Zheng Pan, Shaopan Xiong, Siran Yang, Ju Huang, Yan Zhang, Jiamang Wang, Yong Liu, et al. AMAP Agentic Planning Technical Report. *arXiv preprint arXiv:2512.24957*, 2025. URL <https://arxiv.org/abs/2512.24957>.
- [220] Yang Tian, Zhengpeng Shi, Yu Zhou, and Bo Zhao. Beyond Function Calling: Benchmarking Tool-Using Agents under Tool-Environment Unreliability. *arXiv preprint arXiv:2606.25819*, 2026. URL <https://arxiv.org/abs/2606.25819>.
- [221] Patara Trirat, Wonyong Jeong, and Sung Ju Hwang. AutoML-Agent: A Multi-Agent LLM Framework for Full-Pipeline AutoML. *arXiv preprint arXiv:2410.02958*, 2024. URL <https://arxiv.org/abs/2410.02958>.
- [222] Eren Unlu. Structural Embeddings of Tools for Large Language Models. *arXiv preprint arXiv:2308.00447*, 2023. URL <https://arxiv.org/abs/2308.00447>.
- [223] Andreas Varvarigos, Ali Maatouk, Jiasheng Zhang, Ngoc Bui, Jialin Chen, Leandros Tassioulas, and Rex Ying. LitBench: A Graph-Centric Large Language Model Benchmarking Tool For Literature Tasks. *arXiv preprint arXiv:2603.00051*, 2026. URL <https://arxiv.org/abs/2603.00051>.
- [224] Krithik Vishwanath, Mrigayu Ghosh, Anton Alyakin, Daniel Alexander Alber, Yindalon Aphinyanaphongs, and Eric Karl Oermann. Generalist Large Language Models Outperform Clinical Tools on Medical Benchmarks. *arXiv preprint arXiv:2512.01191*, 2025. URL <https://arxiv.org/abs/2512.01191>.
- [225] Chenyu Wang, Weixin Luo, Sixun Dong, Xiaohua Xuan, Zhengxin Li, Lin Ma, and Shenghua Gao. MLLM-Tool: A Multimodal Large Language Model For Tool Agent Learning. *arXiv preprint arXiv:2401.10727*, 2024. URL <https://arxiv.org/abs/2401.10727>.
- [226] Hanyu Wang, Yuanpu Cao, Lu Lin, and Jinghui Chen. Preflect: From Retrospective to Prospective Reflection in Large Language Model Agents. *arXiv preprint arXiv:2602.07187*, 2026. URL <https://arxiv.org/abs/2602.07187>.
- [227] Junyang Wang, Haiyang Xu, Jiabo Ye, Ming Yan, Weizhou Shen, Ji Zhang, Fei Huang, and Jitao Sang. Mobile-Agent: Autonomous Multimodal Mobile Device Agent with Visual Perception. *arXiv preprint arXiv:2401.16158*, 2024. URL <https://arxiv.org/abs/2401.16158>.
- [228] Kaixin Wang, Tianlin Li, Xiaoyu Zhang, Chong Wang, Weisong Sun, Yang Liu, Aishan Liu, Xiangleong Liu, Chao Shen, and Bin Shi. Software Development Life Cycle Perspective: A Survey of Benchmarks for Code Large Language Models and Agents. *arXiv preprint arXiv:2505.05283*, 2025. URL <https://arxiv.org/abs/2505.05283>.
- [229] Lei Wang, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, Jingsen Zhang, Zhiyuan Chen, Jiakai Tang, Xu Chen, Yankai Lin, Wayne Xin Zhao, Zhewei Wei, and Ji-Rong Wen. A Survey on Large Language Model based Autonomous Agents. *arXiv preprint arXiv:2308.11432*, 2023. URL <https://arxiv.org/abs/2308.11432>.
- [230] Pei Wang, Yanan Wu, Zekun Wang, Jiaheng Liu, Xiaoshuai Song, Zhongyuan Peng, Ken Deng, Chenchen Zhang, Jiakai Wang, Junran Peng, Ge Zhang, Hangyu Guo, Zhaoxiang Zhang, Wenbo Su, and Bo Zheng. MTU-Bench: A Multi-granularity Tool-Use Benchmark for Large Language Models. *arXiv preprint arXiv:2410.11710*, 2024. URL <https://arxiv.org/abs/2410.11710>.
- [231] Qineng Wang, Zihao Wang, Ying Su, and Yangqiu Song. On the Discussion of Large Language Models: Symmetry of Agents and Interplay with Prompts. *arXiv preprint arXiv:2311.07076*, 2023. URL <https://arxiv.org/abs/2311.07076>.
- [232] Renxi Wang, Haonan Li, Xudong Han, Yixuan Zhang, and Timothy Baldwin. Learning From Failure: Integrating Negative Examples when Fine-tuning Large Language Models as Agents. *arXiv preprint arXiv:2402.11651*, 2024. URL <https://arxiv.org/abs/2402.11651>.
- [233] Xi Wang, Hongliang Dai, Shen Gao, and Piji Li. Characteristic AI Agents via Large Language Mod-

- els. *arXiv preprint arXiv:2403.12368*, 2024. URL <https://arxiv.org/abs/2403.12368>.
- [234] Xiaohan Wang, Dian Li, Yilin Zhao, Sinbadliu, and Hui Wang. MetaTool: Facilitating Large Language Models to Master Tools with Meta-task Augmentation. *arXiv preprint arXiv:2407.12871*, 2024. URL <https://arxiv.org/abs/2407.12871>.
- [235] Xiaohan Wang, Yuhui Zhang, Orr Zohar, and Serena Yeung-Levy. VideoAgent: Long-form Video Understanding with Large Language Model as Agent. *arXiv preprint arXiv:2403.10517*, 2024. URL <https://arxiv.org/abs/2403.10517>.
- [236] Yinghao Wang and Cheng Wang. The Observability Gap: Why Output-Level Human Feedback Fails for LLM Coding Agents. *arXiv preprint arXiv:2603.26942*, 2026. URL <https://arxiv.org/abs/2603.26942>.
- [237] Zefan Wang, Zichuan Liu, Yingying Zhang, Aoxiao Zhong, Jihong Wang, Fengbin Yin, Lunting Fan, Lingfei Wu, and Qingsong Wen. RCAGENT: Cloud Root Cause Analysis by Autonomous Agents with Tool-Augmented Large Language Models. *arXiv preprint arXiv:2310.16340*, 2023. URL <https://arxiv.org/abs/2310.16340>.
- [238] Zizhao Wang, Dingcheng Li, Vaishakh Keshava, Phillip Wallis, Ananth Balashankar, Peter Stone, and Lukas Rutishauser. Adversarial Reinforcement Learning for Large Language Model Agent Safety. *arXiv preprint arXiv:2510.05442*, 2025. URL <https://arxiv.org/abs/2510.05442>.
- [239] Lei Wei, Xiao Peng, Jinpeng Ou, and Bin Wang. Think-Augmented Function Calling: Improving LLM Parameter Accuracy Through Embedded Reasoning. *arXiv preprint arXiv:2601.18282*, 2026. URL <https://arxiv.org/abs/2601.18282>.
- [240] Tianxin Wei, Ting-Wei Li, Zhining Liu, Xuying Ning, Ze Yang, Jiaru Zou, Zhichen Zeng, Ruizhong Qiu, Xiao Lin, Dongqi Fu, Zihao Li, Mengting Ai, Duo Zhou, Wenxuan Bao, Yunzhe Li, Gaotang Li, Cheng Qian, Yu Wang, Xiangru Tang, Yin Xiao, et al. Agentic Reasoning for Large Language Models. *arXiv preprint arXiv:2601.12538*, 2026. URL <https://arxiv.org/abs/2601.12538>.
- [241] Duo Wu, Jinghe Wang, Yuan Meng, Yanning Zhang, Le Sun, and Zhi Wang. CATP-LLM: Empowering Large Language Models for Cost-Aware Tool Planning. *arXiv preprint arXiv:2411.16313*, 2024. URL <https://arxiv.org/abs/2411.16313>.
- [242] Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yiwen Ding, Boyang Hong, Ming Zhang, Junzhe Wang, Senjie Jin, Enyu Zhou, Rui Zheng, Xiaoran Fan, Xiao Wang, Limao Xiong, Yuhao Zhou, Weiran Wang, Changhao Jiang, Yicheng Zou, Xiangyang Liu, Zhangyue Yin, et al. The Rise and Potential of Large Language Model Based Agents: A Survey. *arXiv preprint arXiv:2309.07864*, 2023. URL <https://arxiv.org/abs/2309.07864>.
- [243] Meng Xiao, Xunxin Cai, Qingqing Long, Chengrui Wang, Yuanchun Zhou, and Hengshu Zhu. Knowledge-Driven Agentic Scientific Corpus Distillation Framework for Biomedical Large Language Models Training. *arXiv preprint arXiv:2504.19565*, 2025. URL <https://arxiv.org/abs/2504.19565>.
- [244] Rui Xiao, Jiong Wang, Lu Han, Na Zong, and Han Wu. Molly: Making Large Language Model Agents Solve Python Problem More Logically. *arXiv preprint arXiv:2412.18093*, 2024. URL <https://arxiv.org/abs/2412.18093>.
- [245] Chengxing Xie, Canyu Chen, Feiran Jia, Ziyu Ye, Shiyang Lai, Kai Shu, Jindong Gu, Adel Bibi, Ziniu Hu, David Jurgens, James Evans, Philip Torr, Bernard Ghanem, and Guohao Li. Can Large Language Model Agents Simulate Human Trust Behavior? *arXiv preprint arXiv:2402.04559*, 2024. URL <https://arxiv.org/abs/2402.04559>.
- [246] Hao-Xiang Xu, Chong Deng, Jiaqing Liu, Wen Wang, Qian Chen, Lujia Bao, Xiangang Li, and Zhen-Hua Ling. GenesisFunc: Multi-Agent Data Generation for Accurate and Generalizable Function-Calling. *arXiv preprint arXiv:2605.28835*, 2026. URL <https://arxiv.org/abs/2605.28835>.
- [247] Mengdi Xu, Peide Huang, Wenhao Yu, Shiqi Liu, Xilun Zhang, Yaru Niu, Tingnan Zhang, Fei Xia, Jie Tan, and Ding Zhao. Creative Robot Tool Use with Large Language Models. *arXiv preprint arXiv:2310.13065*, 2023. URL <https://arxiv.org/abs/2310.13065>.
- [248] Qiancheng Xu, Yongqi Li, Heming Xia, and Wenjie Li. Enhancing Tool Retrieval with Iterative Feedback from Large Language Models. *arXiv preprint arXiv:2406.17465*, 2024. URL <https://arxiv.org/abs/2406.17465>.
- [249] Qiancheng Xu, Yongqi Li, Heming Xia, Fan Liu, Min Yang, and Wenjie Li. PEToolLLM: Towards Personalized Tool Learning in Large Language Models. *arXiv preprint arXiv:2502.18980*, 2025. URL <https://arxiv.org/abs/2502.18980>.
- [250] Qiancheng Xu, Yongqi Li, Fan Liu, Hongru Wang, Min Yang, and Wenjie Li. TInR: Exploring Tool-Internalized Reasoning in Large Language Models. *arXiv preprint arXiv:2604.10788*, 2026. URL <https://arxiv.org/abs/2604.10788>.
- [251] Qiantong Xu, Fenglu Hong, Bo Li, Changran Hu, Zhengyu Chen, and Jian Zhang. On the Tool Manipulation Capability of Open-source Large Language Models. *arXiv preprint arXiv:2305.16504*, 2023. URL <https://arxiv.org/abs/2305.16504>.
- [252] Renjun Xu and Yang Yan. Agent Skills for Large

- Language Models: Architecture, Acquisition, Security, and the Path Forward. *arXiv preprint arXiv:2602.12430*, 2026. URL <https://arxiv.org/abs/2602.12430>.
- [253] Zhenran Xu, Senbao Shi, Baotian Hu, Jindi Yu, Dongfang Li, Min Zhang, and Yuxiang Wu. Towards Reasoning in Large Language Models via Multi-Agent Peer Review Collaboration. *arXiv preprint arXiv:2311.08152*, 2023. URL <https://arxiv.org/abs/2311.08152>.
- [254] Dingkan Yang, Jinjie Wei, Mingcheng Li, Jiyao Liu, Lihao Liu, Ming Hu, Junjun He, Yakun Ju, Wei Zhou, Yang Liu, and Lihua Zhang. MedAide: Information Fusion and Anatomy of Medical Intent via LLM-based Agent Collaboration. *arXiv preprint arXiv:2410.12532*, 2024. URL <https://arxiv.org/abs/2410.12532>.
- [255] Ke Yang, Jiateng Liu, John Wu, Chaoqi Yang, Yi R. Fung, Sha Li, Zixuan Huang, Xu Cao, Xingyao Wang, Yiquan Wang, Heng Ji, and Chengxiang Zhai. If LLM Is the Wizard, Then Code Is the Wand: A Survey on How Code Empowers Large Language Models to Serve as Intelligent Agents. *arXiv preprint arXiv:2401.00812*, 2024. URL <https://arxiv.org/abs/2401.00812>.
- [256] Rui Yang, Lin Song, Yanwei Li, Sijie Zhao, Yixiao Ge, Xiu Li, and Ying Shan. GPT4Tools: Teaching Large Language Model to Use Tools via Self-instruction. *arXiv preprint arXiv:2305.18752*, 2023. URL <https://arxiv.org/abs/2305.18752>.
- [257] Seungbin Yang, ChaeHun Park, Taehee Kim, and Jaegul Choo. Can Tool-augmented Large Language Models be Aware of Incomplete Conditions? *arXiv preprint arXiv:2406.12307*, 2024. URL <https://arxiv.org/abs/2406.12307>.
- [258] Shu Yang, Shenzhe Zhu, Hao Zhu, Jose Ramon Enriquez, Di Wang, Alex Pentland, Michiel A. Bakker, and Jiaxin Pei. Multi-User Large Language Model Agents. *arXiv preprint arXiv:2604.08567*, 2026. URL <https://arxiv.org/abs/2604.08567>.
- [259] Shuting Yang, Zehui Liu, and Wolfgang Mayer. ShizishanGPT: An Agricultural Large Language Model Integrating Tools and Resources. *arXiv preprint arXiv:2409.13537*, 2024. URL <https://arxiv.org/abs/2409.13537>.
- [260] Wenkai Yang, Xiaohan Bi, Yankai Lin, Sishuo Chen, Jie Zhou, and Xu Sun. Watch Out for Your Agents! Investigating Backdoor Threats to LLM-Based Agents. *arXiv preprint arXiv:2402.11208*, 2024. URL <https://arxiv.org/abs/2402.11208>.
- [261] Huanjin Yao, Ruifei Zhang, Jiaxing Huang, Jingyi Zhang, Yibo Wang, Bo Fang, Ruolin Zhu, Yongcheng Jing, Shunyu Liu, Guanbin Li, and Dacheng Tao. A Survey on Agentic Multi-modal Large Language Models. *arXiv preprint arXiv:2510.10991*, 2025. URL <https://arxiv.org/abs/2510.10991>.
- [262] Junjie Ye, Guanyu Li, Songyang Gao, Caishuang Huang, Yilong Wu, Sixian Li, Xiaoran Fan, Shihan Dou, Tao Ji, Qi Zhang, Tao Gui, and Xuanjing Huang. ToolEyes: Fine-Grained Evaluation for Tool Learning Capabilities of Large Language Models in Real-world Scenarios. *arXiv preprint arXiv:2401.00741*, 2024. URL <https://arxiv.org/abs/2401.00741>.
- [263] Junjie Ye, Sixian Li, Guanyu Li, Caishuang Huang, Songyang Gao, Yilong Wu, Qi Zhang, Tao Gui, and Xuanjing Huang. ToolSword: Unveiling Safety Issues of Large Language Models in Tool Learning Across Three Stages. *arXiv preprint arXiv:2402.10753*, 2024. URL <https://arxiv.org/abs/2402.10753>.
- [264] Junjie Ye, Yilong Wu, Songyang Gao, Caishuang Huang, Sixian Li, Guanyu Li, Xiaoran Fan, Qi Zhang, Tao Gui, and Xuanjing Huang. RoT-Bench: A Multi-Level Benchmark for Evaluating the Robustness of Large Language Models in Tool Learning. *arXiv preprint arXiv:2401.08326*, 2024. URL <https://arxiv.org/abs/2401.08326>.
- [265] Junjie Ye, Yilong Wu, Sixian Li, Yuming Yang, Zhiheng Xi, Tao Gui, Qi Zhang, Xuanjing Huang, Peng Wang, Zhongchao Shi, Jianping Fan, and Zhengyin Du. TL-Training: A Task-Feature-Based Framework for Training Large Language Models in Tool Use. *arXiv preprint arXiv:2412.15495*, 2024. URL <https://arxiv.org/abs/2412.15495>.
- [266] Junjie Ye, Zhengyin Du, Xuesong Yao, Weijian Lin, Yufei Xu, Zehui Chen, Zaiyuan Wang, Sining Zhu, Zhiheng Xi, Siyu Yuan, Tao Gui, Qi Zhang, Xuanjing Huang, and Jiecao Chen. ToolHop: A Query-Driven Benchmark for Evaluating Large Language Models in Multi-Hop Tool Use. *arXiv preprint arXiv:2501.02506*, 2025. URL <https://arxiv.org/abs/2501.02506>.
- [267] Junjie Ye, Changhao Jiang, Zhengyin Du, Yufei Xu, Xuesong Yao, Zhiheng Xi, Xiaoran Fan, Qi Zhang, Tao Gui, Xuanjing Huang, and Jiecao Chen. Feedback-Driven Tool-Use Improvements in Large Language Models via Automated Build Environments. *arXiv preprint arXiv:2508.08791*, 2025. URL <https://arxiv.org/abs/2508.08791>.
- [268] Yaoqi Ye, Yiran Zhao, Keyu Duan, Zeyu Zheng, Kenji Kawaguchi, Cihang Xie, and Michael Qizhe Shieh. In-Context Reinforcement Learning for Tool Use in Large Language Models. *arXiv preprint arXiv:2603.08068*, 2026. URL <https://arxiv.org/abs/2603.08068>.
- [269] Fan Yin, Zifeng Wang, I-Hung Hsu, Jun Yan,

- Ke Jiang, Yanfei Chen, Jindong Gu, Long T. Le, Kai-Wei Chang, Chen-Yu Lee, Hamid Palangi, and Tomas Pfister. Magnet: Multi-turn Tool-use Data Synthesis and Distillation via Graph Translation. *arXiv preprint arXiv:2503.07826*, 2025. URL <https://arxiv.org/abs/2503.07826>.
- [270] Shuo Yin, Weihao You, Zhilong Ji, Guoqiang Zhong, and Jinfeng Bai. MuMath-Code: Combining Tool-Use Large Language Models with Multi-perspective Data Augmentation for Mathematical Reasoning. *arXiv preprint arXiv:2405.07551*, 2024. URL <https://arxiv.org/abs/2405.07551>.
- [271] Luke Yoffe, Alfonso Amayuelas, and William Yang Wang. DebUnc: Improving Large Language Model Agent Communication With Uncertainty Metrics. *arXiv preprint arXiv:2407.06426*, 2024. URL <https://arxiv.org/abs/2407.06426>.
- [272] Yi Yu, Liuyi Yao, Yuexiang Xie, Qingquan Tan, Jiaqi Feng, Yaliang Li, and Libing Wu. Agentic Memory: Learning Unified Long-Term and Short-Term Memory Management for Large Language Model Agents. *arXiv preprint arXiv:2601.01885*, 2026. URL <https://arxiv.org/abs/2601.01885>.
- [273] Murong Yue. A Survey of Large Language Model Agents for Question Answering. *arXiv preprint arXiv:2503.19213*, 2025. URL <https://arxiv.org/abs/2503.19213>.
- [274] Xin Zeng, Xiaoyu Wang, Tengxiang Zhang, Chun Yu, Shengdong Zhao, and Yiqiang Chen. GestureGPT: Toward Zero-Shot Free-Form Hand Gesture Understanding with Large Language Model Agents. *arXiv preprint arXiv:2310.12821*, 2023. URL <https://arxiv.org/abs/2310.12821>.
- [275] Bang Zhang, Ruotian Ma, Qingxuan Jiang, Peisong Wang, Jiaqi Chen, Zheng Xie, Xingyu Chen, Yue Wang, Fanghua Ye, Jian Li, Yifan Yang, Zhaopeng Tu, and Xiaolong Li. Sentient Agent as a Judge: Evaluating Higher-Order Social Cognition in Large Language Models. *arXiv preprint arXiv:2505.02847*, 2025. URL <https://arxiv.org/abs/2505.02847>.
- [276] Ke Zhang, Xiaoning Zhao, Ce Zheng, Jiahong Ning, Dandan Zhu, Wenqi Zhang, Chen Sun, and Toshiharu Sugawara. Tool-RoCo: An Agent-as-Tool Self-organization Large Language Model Benchmark in Multi-robot Cooperation. *arXiv preprint arXiv:2511.21510*, 2025. URL <https://arxiv.org/abs/2511.21510>.
- [277] Minxing Zhang, Yi Yang, Roy Xie, Bhuwan Dhingra, Shuyan Zhou, and Jian Pei. Generalizability of Large Language Model-Based Agents: A Comprehensive Survey. *arXiv preprint arXiv:2509.16330*, 2025. URL <https://arxiv.org/abs/2509.16330>.
- [278] Ruixin Zhang, Jon Donnelly, Zhicheng Guo, Ghazal Khalighinejad, Haiyang Huang, Alina Jade Barnett, and Cynthia Rudin. It’s LIT! Reliability-Optimized LLMs with Inspectable Tools. *arXiv preprint arXiv:2511.14903*, 2025. URL <https://arxiv.org/abs/2511.14903>.
- [279] Wei Zhang, Yi Zhang, Li Zhu, Qianghuai Jia, Feijun Jiang, Hongcheng Guo, Zhoujun Li, and Mengping Zhou. ADC: Enhancing Function Calling Via Adversarial Datasets and Code Line-Level Feedback. *arXiv preprint arXiv:2412.17754*, 2024. URL <https://arxiv.org/abs/2412.17754>.
- [280] Xijia Zhang, Yue Guo, Simon Stepputtis, Katia Sycara, and Joseph Campbell. Explaining Agent Behavior with Large Language Models. *arXiv preprint arXiv:2309.10346*, 2023. URL <https://arxiv.org/abs/2309.10346>.
- [281] Yinger Zhang, Hui Cai, Xeirui Song, Yicheng Chen, Rui Sun, and Jing Zheng. Reverse Chain: A Generic-Rule for LLMs to Master Multi-API Planning. *arXiv preprint arXiv:2310.04474*, 2023. URL <https://arxiv.org/abs/2310.04474>.
- [282] Yuxiang Zhang, Jing Chen, Junjie Wang, Yaxin Liu, Cheng Yang, Chufan Shi, Xinyu Zhu, Zihao Lin, Hanwen Wan, Yujiu Yang, Tetsuya Sakai, Tian Feng, and Hayato Yamana. ToolBeHonest: A Multi-level Hallucination Diagnostic Benchmark for Tool-Augmented Large Language Models. *arXiv preprint arXiv:2406.20015*, 2024. URL <https://arxiv.org/abs/2406.20015>.
- [283] Zeyi Zhang, Yanju Zhou, Heyuan Yao, Tenglong Ao, Xiaohang Zhan, and Libin Liu. Social Agent: Mastering Dyadic Nonverbal Behavior Generation via Conversational LLM Agents. *arXiv preprint arXiv:2510.04637*, 2025. URL <https://arxiv.org/abs/2510.04637>.
- [284] Haiteng Zhao, Chang Ma, Guoyin Wang, Jing Su, Lingpeng Kong, Jingjing Xu, Zhi-Hong Deng, and Hongxia Yang. Empowering Large Language Model Agents through Action Learning. *arXiv preprint arXiv:2402.15809*, 2024. URL <https://arxiv.org/abs/2402.15809>.
- [285] Pengyu Zhao, Zijian Jin, and Ning Cheng. An In-depth Survey of Large Language Model-based Artificial Intelligence Agents. *arXiv preprint arXiv:2309.14365*, 2023. URL <https://arxiv.org/abs/2309.14365>.
- [286] Weixiang Zhao, Yingshuo Wang, Yichen Zhang, Yang Deng, Yanyan Zhao, Wanxiang Che, Bing Qin, and Ting Liu. Large Language Model Agents Are Not Always Faithful Self-Evolvers. *arXiv preprint arXiv:2601.22436*, 2026. URL <https://arxiv.org/abs/2601.22436>.
- [287] Zhen Zhao, Dunbing Tang, Changchun Liu, Liping Wang, Zequn Zhang, Haihua Zhu, Kai Chen, Qingwei Nie, and Yuchen Ji. A Large Language Model-based multi-agent manufacturing system for intel-

- lignent shopfloor. *arXiv preprint arXiv:2405.16887*, 2024. URL <https://arxiv.org/abs/2405.16887>.
- [288] Lucen Zhong, Zhengxiao Du, Xiaohan Zhang, Haiyi Hu, and Jie Tang. ComplexFuncBench: Exploring Multi-Step and Constrained Function Calling under Long-Context Scenario. *arXiv preprint arXiv:2501.10132*, 2025. URL <https://arxiv.org/abs/2501.10132>.
- [289] Jiawei Zhou, Amy Z. Chen, Darshi Shah, Laura M. Schwab Reese, and Munmun De Choudhury. A Risk Taxonomy and Reflection Tool for Large Language Model Adoption in Public Health. *arXiv preprint arXiv:2411.02594*, 2024. URL <https://arxiv.org/abs/2411.02594>.
- [290] Wangchunshu Zhou, Yuchen Eleanor Jiang, Long Li, Jialong Wu, Tiannan Wang, Shi Qiu, Jintian Zhang, Jing Chen, Ruipu Wu, Shuai Wang, Shiding Zhu, Jiyu Chen, Wentao Zhang, Xiangru Tang, Ningyu Zhang, Huajun Chen, Peng Cui, and Mrinmaya Sachan. Agents: An Open-source Framework for Autonomous Language Agents. *arXiv preprint arXiv:2309.07870*, 2023. URL <https://arxiv.org/abs/2309.07870>.
- [291] Wangchunshu Zhou, Yixin Ou, Shengwei Ding, Long Li, Jialong Wu, Tiannan Wang, Jiamin Chen, Shuai Wang, Xiaohua Xu, Ningyu Zhang, Huajun Chen, and Yuchen Eleanor Jiang. Symbolic Learning Enables Self-Evolving Agents. *arXiv preprint arXiv:2406.18532*, 2024. URL <https://arxiv.org/abs/2406.18532>.
- [292] Wei Zhou, Xuanhe Zhou, Shaokun Han, Hongming Xu, Guoliang Li, Zhiyu Li, Feiyu Xiong, and Fan Wu. Are We Ready For An Agent-Native Memory System? *arXiv preprint arXiv:2606.24775*, 2026. URL <https://arxiv.org/abs/2606.24775>.
- [293] Weijie Zhou, Manli Tao, Chaoyang Zhao, Honghui Dong, Ming Tang, and Jinqiao Wang. LightPlanner: Unleashing the Reasoning Capabilities of Lightweight Large Language Models in Task Planning. *arXiv preprint arXiv:2503.08508*, 2025. URL <https://arxiv.org/abs/2503.08508>.
- [294] Zihao Zhou, Bin Hu, Chenyang Zhao, Pu Zhang, and Bin Liu. Large Language Model as a Policy Teacher for Training Reinforcement Learning Agents. *arXiv preprint arXiv:2311.13373*, 2023. URL <https://arxiv.org/abs/2311.13373>.
- [295] Feiyu Zhu and Reid Simmons. Bootstrapping Cognitive Agents with a Large Language Model. *arXiv preprint arXiv:2403.00810*, 2024. URL <https://arxiv.org/abs/2403.00810>.
- [296] Kaijie Zhu, Jindong Wang, Qinlin Zhao, Ruochen Xu, and Xing Xie. Dynamic Evaluation of Large Language Models by Meta Probing Agents. *arXiv preprint arXiv:2402.14865*, 2024. URL <https://arxiv.org/abs/2402.14865>.
- [297] Longtao Zhu, Hongyu Yang, Ge Song, Xin Ma, Yanxin Zhang, and Yulong Ji. AutoFPDesigner: Automated Flight Procedure Design Based on Multi-Agent Large Language Model. *arXiv preprint arXiv:2410.14989*, 2024. URL <https://arxiv.org/abs/2410.14989>.
- [298] Yuchen Zhuang, Xiang Chen, Tong Yu, Saayan Mitra, Victor Bursztyn, Ryan A. Rossi, Somdeb Sarkhel, and Chao Zhang. ToolChain*: Efficient Action Space Navigation in Large Language Models with A* Search. *arXiv preprint arXiv:2310.13227*, 2023. URL <https://arxiv.org/abs/2310.13227>.
- [299] Yuchen Zhuang, Jingfeng Yang, Haoming Jiang, Xin Liu, Kewei Cheng, Sanket Lokegaonkar, Yifan Gao, Qing Ping, Tianyi Liu, Binxuan Huang, Zheng Li, Zhengyang Wang, Pei Chen, Ruijie Wang, Rongzhi Zhang, Nasser Zalmout, Priyanka Nigam, Bing Yin, and Chao Zhang. Hephaestus: Improving Fundamental Agent Capabilities of Large Language Models through Continual Pre-Training. *arXiv preprint arXiv:2502.06589*, 2025. URL <https://arxiv.org/abs/2502.06589>.
- [300] Terry Yue Zhuo, Minh Chien Vu, Jenny Chim, Han Hu, Wenhao Yu, Ratnadira Widayarsi, Imam Nur Bani Yusuf, Haolan Zhan, Junda He, Indraneil Paul, Simon Brunner, Chen Gong, Thong Hoang, Armel Randy Zebaze, Xiaoheng Hong, Wen-Ding Li, Jean Kaddour, Ming Xu, Zhihan Zhang, Prateek Yadav, et al. BigCodeBench: Benchmarking Code Generation with Diverse Function Calls and Complex Instructions. *arXiv preprint arXiv:2406.15877*, 2024. URL <https://arxiv.org/abs/2406.15877>.